

# Supporting Information

## Topological Systems with CP2K

Thomas D. Kühne, Alexander Cerjan, Vladislav Efremkin,  
Hermann Schulz-Baldes, and Emil Prodan

Corresponding author: [tkuehne@cp2k.org](mailto:tkuehne@cp2k.org)

### S1 Scope and Reproduction Boundaries

This working document combines the retained band-topology calculations transferred from the k-point manuscript with separately validated transport, localizer, and quadratic-pseudospectrum developments. The original results are not recomputed or relabelled by moving the manuscript text. The development status is recorded as of 26 September 2026. A source freeze and complete public reproduction archive remain submission requirements.

#### Applications versus test calculations

TopoHSE-DB and the submitted generative-predictive discovery study are cited in the main text as materials-science applications. Their VASP-based calculations are not part of the CP2K validation records collected here. In particular, their database labels and candidate counts do not validate the new CP2K localizer or transport implementations.

The smaller calculations below are tests and demonstrations. The companion k-point paper additionally reports 18 converged Si/Al calculations in conventional cubic cells with eight Si atoms ( $a = 5.43$  Å) and four Al atoms ( $a = 4.05$  Å). These use PBE, DZVP-MOLOPT-SR-GTH bases, GTH-PBE pseudopotentials, 600/60 Ry grid cutoffs, and an SCF threshold of  $10^{-9}$ . Al uses Fermi-Dirac occupations at 1000 K, so its compared quantity is the electronic free energy. The same-mesh comparison uses  $4 \times 4 \times 4$  sampling: 64 full-grid points, 32 with time reversal, and four with either K290 or SPGLIB atomic symmetry. The maximum cell-energy/free-energy difference is  $1.4 \times 10^{-12}$  hartree; the Al atomic-symmetry stress difference is 0.098 bar. These tests demonstrate representation consistency, not independently converged material properties or general force accuracy. Their source and raw records remain those of the companion manuscript; no new Si/Al calculations were performed for the present discussion.

### S2 Native Wilson-Loop Test and Demonstration Calculations

The topology examples form a separate calculation set from the Si/Al mainline tests. The native extension supplies directed Gaussian-basis overlaps, second-variational spinors, and the Wilson-loop analysis described in the main text. The retained calculations were completed on 17 September 2026 with one MPI process and one OpenMP thread. Source provenance and artifact checksums are recorded in `validation/topology/README.md` and the figure manifest.

Both examples use PBE, the DZVP-MOLOPT-GGA-GTH basis from `BASIS_MOLOPT_UZH`, SOC pseudopotentials from `GTH_SOC_POTENTIALS`, a 200 Ry density cutoff and a 40 Ry relative cutoff. The potential labels are `GTH-PBE-q8` for neon and `GTH-PBE-q4` for tin. Complex orbitals and explicit full SCF meshes are used. The neon reference contains one atom at fractional coordinate  $(1/2, 1/2, 1/2)$  in a cubic cell of side 5 Å, with a  $2 \times 2 \times 2$  SCF mesh and convergence threshold  $10^{-9}$ . The stanene cell has lattice vectors  $(4.674, 0, 0)$ ,  $(2.337, 4.047802737288466, 0)$ , and  $(0, 0, 20)$  Å, and two tin atoms at fractional positions  $(1/3, 1/3, 0.5213)$  and  $(2/3, 2/3, 0.4787)$ . It therefore

has a buckling height of 0.852 Å. These fixed geometric parameters follow the pristine-stanene reference reported by Zhang et al.<sup>1</sup> An  $8 \times 8 \times 1$  SCF mesh and convergence threshold  $10^{-8}$  are used for stanene. The examples test the topology implementation at these fixed settings, rather than constitute a new structural or band-gap prediction.

Eight occupied spinors are retained in each case. The stanene second-variational calculation includes all 26 scalar basis states, giving 52 spinor states before occupied-band selection. For neon, five scalar states produce ten spinors. The selected half-plane is  $\mathbf{k}(s, t) = (t, s/2, 0)$  in fractional reciprocal coordinates, with  $0 \leq s, t \leq 1$ . The loop closes with reciprocal shift  $(1, 0, 0)$ . Starting meshes contain four points on each of three loops for neon and 12 points on each of 13 loops for stanene. Refinement doubles the points per loop and inserts a midpoint between neighboring transverse loops. The accepted surfaces have  $8 \times 5$  and  $192 \times 193$  points, respectively. The last center changes are  $1.28 \times 10^{-5}$  and  $2.73 \times 10^{-4}$  in reduced units. The smallest link singular values are 0.99826 and 0.95248.

The native invariants are zero for neon and one for stanene. Z2Pack 2.2.1 gives the same stanene parity using either the raw overlaps or their unitary polar factors. For the latter, the maximum Wilson-eigenvalue difference from the native calculation is  $7.95 \times 10^{-14}$ . This comparison tests the loop algebra with identical input matrices. Separate operator checks enforce zero-step normalization, reverse-link adjoint relations, loop reversal, and atomic-position phases. A bonded two-atom full-spinor check gives maximum adjoint and zero-step residuals of  $2.79 \times 10^{-14}$  and  $2.00 \times 10^{-13}$ , respectively.

An additional adaptive Z2Pack-driven stanene surface uses 42 CP2K line calculations and ends with 15 accepted transverse lines, all satisfying the line-position, movement, and gap checks, and again gives  $\nu = 1$ . This calculation chooses its sampling independently but uses the same CP2K Hamiltonian and overlap implementation. The main-text figure shows the native samples without interpolation or artificial separation of degenerate centers. It is regenerated by `python3 validation/topology/plot_wilson.py`, which requires NumPy and Matplotlib. Raw `.wilson` files, inputs, logs, comparison reports, generated comma-separated figure data, and checksums accompany the figure.

## S3 Native Chern and Cross-Geometry Extension

### S3.1 Source versions and independent analysis routes

The Chern and cross-geometry checks were completed on 18 September 2026 with a separate development executable. They are distinct from the retained 17 September neon/stanene calculations and do not reassign those earlier results to a newer implementation. Source patches, executable records, library versions, and independent comparison scripts accompany the validation archive. The numerical comparison tools require NumPy and SciPy; Z2Pack 2.2.1 is an optional independent analysis dependency.

Table S1: Topology analysis routes and runtime requirements. All native routes act on Gaussian-basis electronic states without constructing maximally localized Wannier orbitals.

| Route                         | Quantity and sampling                                                                           | Dependencies and scope                                                                            |
|-------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Native Wilson/ $\mathbb{Z}_2$ | Wilson spectra, Berry phases, and time-reversal half-plane $\mathbb{Z}_2$ .                     | CP2K native kernels, including second-variational SOC. No Z2Pack or Wannier90 runtime dependency. |
| Native Wilson/ $C_1$          | Determinant-phase winding on a full closed reciprocal surface, selected by <code>CHERN</code> . | CP2K native kernels. Fixed-rank isolated subspace, closure and refinement checks.                 |

<sup>1</sup>R.-w. Zhang et al., Room temperature quantum spin Hall insulator in ethynyl-derivative functionalized stanene films, *Sci. Rep.* **6**, 18879 (2016), <https://doi.org/10.1038/srep18879>.

| Route                          | Quantity and sampling                                                                                                            | Dependencies and scope                                                                                                                                     |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Native inversion analysis      | Even/odd irreps, Fu–Kane indices and inversion-subgroup EBR signatures at TRIM.                                                  | Separate native extension, Section S4. No general space-group catalogue.                                                                                   |
| Native parameter-mesh analysis | $C_1$ on two parameters and the second Chern character pairing $C_2$ on four parameters, using physical cross-geometry overlaps. | <code>topology_phasons</code> reads <code>STATE_EXPORT</code> files. One process with optional OpenMP. No Python, Z2Pack, or Wannier90 runtime dependency. |
| External Z2Pack route          | Independent loop/surface analysis of CP2K overlaps, with optional adaptive CP2K sampling.                                        | Python and Z2Pack. Optional analysis and cross-validation, not the implementation of the native invariants.                                                |

### S3.2 Parameter-mesh discretization and physical closure

Let  $Q_\mu(x)$  be the polar-unitary selected-state overlap directed from mesh vertex  $x$  to  $x + \hat{\mu}$ . All boundary links include the specified physical sewing. The oriented plaquette is

$$P_{\mu\nu}(x) = Q_\mu(x)Q_\nu(x + \hat{\mu})Q_\mu^\dagger(x + \hat{\nu})Q_\nu^\dagger(x). \quad (\text{S1})$$

With this link orientation, the two-parameter estimator is

$$C_1^{\text{mesh}} = -\frac{1}{2\pi} \sum_x \text{Arg det } P_{12}(x). \quad (\text{S2})$$

Its sign matches the determinant Wilson-phase winding convention of the main text. For four parameters, define the full principal matrix logarithm  $F_{\mu\nu}(x) = \log P_{\mu\nu}(x)$ . Each  $F_{\mu\nu}$  is anti-Hermitian and already includes the plaquette area. The estimator is

$$C_2^{\text{mesh}} = -\frac{1}{4\pi^2} \sum_x \text{Re Tr } [F_{12}F_{34} - F_{13}F_{24} + F_{14}F_{23}](x). \quad (\text{S3})$$

The full matrices are needed for a non-Abelian occupied subspace. Equation (S3) approximates the second Chern character pairing and is not generally the second Chern class. Its finite-mesh value is reported without integer rounding.

Physical links use  $C_a^\dagger O_{ab} C_b$ , with both spinor components summed when present. The cross-geometry AO matrix  $O_{ab}$  retains the moving contracted Gaussian centers, periodic images, and reciprocal boundary phases. The cell, atom count and selected rank remain fixed throughout the family. Endpoint checks use explicit atom permutations, lattice translations, basis matching, and selected-subspace sewing. The code reports AO-metric errors, sampled selected/excluded-band gaps, minimum link singular values, and principal plaquette phases. A refined mesh must preserve the same physical family. Sampled isolation and invariant convergence do not by themselves establish a bulk spectral gap between samples. An indirect gap is interpreted only for a lowest-band prefix with a justified common energy reference across geometries.

### S3.3 Retained validation results

The extension was rebuilt with GNU Fortran 16 and OpenBLAS. Its curvature and snapshot unit tests and the existing Wilson/ $C_1/\mathbb{Z}_2$  unit test passed with two OpenMP threads. All four inputs in `QS/regtest-topology` passed with two MPI ranks and two OpenMP threads per rank. The auxiliary parameter-mesh executable itself was tested as one process, not as an MPI-distributed postprocessor.

The native tests cover singular or underresolved links, metric and seam failures, determinant phases, non-Abelian gauge covariance, and model refinement. For the occupied doublet of the

four-dimensional Dirac model at mass  $-3$ , the  $8^4$  estimate is  $-0.7424128463$  and is closer to the continuum value  $-1$  than the  $6^4$  result. This is a finite-resolution regression check, not a claim of a converged integer at  $8^4$ . Independent Python/native comparisons additionally check gauge changes and reversal of parameter orientation.

For archived CP2K state exports covering skew cells, multiple centers, SOC spinors and reciprocal-boundary links, the largest difference between native and independently evaluated Gaussian overlaps is  $1.03 \times 10^{-15}$  with either one or two OpenMP threads. A cross-geometry overlap agrees to  $5.56 \times 10^{-16}$ . Table S2 reports physical atomic-translation pumps. The two-parameter case varies one reciprocal coordinate and one periodic atomic displacement. The separable four-parameter case uses the ordered coordinates  $(k_x, k_y, \phi_x, \phi_y)$ .

Table S2: Native physical Gaussian-snapshot tests. The minimum singular value is measured before polar unitarization. The phase column reports the largest diagnostic plaquette phase in radians. The  $4^4$  separable case tests the four-parameter physical workflow, not generic  $C_2$  convergence.

| Mesh           | Unrounded pairing     | Minimum singular value | Maximum phase |
|----------------|-----------------------|------------------------|---------------|
| $4 \times 4$   | $C_1 = -1.0000000000$ | 0.234278               | 0.402344      |
| $8 \times 8$   | $C_1 = -1.0000000000$ | 0.609933               | 0.099083      |
| $12 \times 12$ | $C_1 = -1.0000000000$ | 0.780474               | 0.044040      |
| $4^4$          | $C_2 = -1.0000000000$ | 0.233877               | 0.402347      |

The native  $4 \times 4$  to  $8 \times 8$  physical refinement check passed at an invariant tolerance of  $10^{-8}$ . The independent comparison-tool suite passed 34 tests without Z2Pack, with five optional-adaptor tests skipped, and all 39 tests with Z2Pack installed. The accompanying `validation/topology/native-phasons/` directory contains the machine-readable overlap and pump results, test logs, and source-version record. These are implementation tests and introduce no new material prediction or change to the neon/stanene results.

## S4 Native Inversion Representations and Symmetry Signatures

The inversion extension was validated separately from the phason extension and from the earlier Si/Al and Wilson calculations. Its source patch, base snapshot, inputs and outputs remain in `validation/topology/inversion-tqc/`. These records identify the original tests without retrospectively assigning them to another executable.

The WANNIER90 print section accepts KPOINTS\_SOURCE TRIM, SOC T, TIME\_REVERSAL T and INVERSION\_TQC T. TQC\_DIMENSION selects the four  $k_z = 0$  TRIM or all eight three-dimensional TRIM. PARITY\_ORIGIN specifies the inversion center in fractional direct coordinates. The occupied spinor bands must be a complete lowest-band prefix, with excluded states available for checking spectral separation. A PARITY-only mode also accepts isolated scalar subspaces. Neither route requires Z2Pack, Wannier90 execution or a Wannier fit. The TRIM calculation holds the converged SCF potential fixed and does not require the SCF integration mesh to contain those points. The two-dimensional option samples the fractional  $k_z = 0$  plane of a three-dimensionally periodic cell, not a separate two-dimensional Poisson boundary condition. The inversion center is specified explicitly rather than inferred from a space-group search.

For an orbital of angular momentum  $l$  on atom  $a$ , inversion about  $\mathbf{c}$  maps it to an identical Gaussian orbital on atom  $b$  when  $2\mathbf{c} - \mathbf{r}_a = \mathbf{r}_b + \mathbf{L}$  with integer  $\mathbf{L}$ . The coefficient multiplier is  $(-1)^l \exp(2\pi i \mathbf{k} \cdot \mathbf{L})$ . The code verifies atom-kind matching, involution, metric covariance,  $C^\dagger S C = I$ , unitarity and Hermiticity of the projected inversion matrix, and its commutator with the selected eigenvalues. Spinful time reversal is checked through the sewing matrix of  $\Theta = i\sigma_y K$ . Parity multiplicities must be even before division into Kramers pairs. The dimensionless and geometry tolerances are  $10^{-6}$  and  $10^{-6}$  bohr, respectively. The energy-commutator tolerance is

$10^{-6}$  hartree.

For  $N$  occupied Kramers pairs and odd counts  $n_{\kappa}^{-}$  at  $\mathbf{k} = \kappa/2$ ,  $\kappa \in \{0, 1\}^d$ , the parity index is  $\nu = \sum_{\kappa} n_{\kappa}^{-} \bmod 2$ . In three dimensions the weak indices restrict this sum to  $\kappa_i = 1$ , and the convention for the inversion indicator is  $z_4 = \sum_{\kappa} n_{\kappa}^{-} \bmod 4$ . The analytic inversion-subgroup EBRs are local even or odd Kramers pairs at centers  $\mathbf{c} + \mathbf{a}/2$ . Define

$$d_{\mathbf{a}} = 2^{-d} \sum_{\kappa} (-1)^{\mathbf{a} \cdot \kappa} (N - 2n_{\kappa}^{-}). \quad (\text{S4})$$

An integer atomic-signature decomposition exists when all  $d_{\mathbf{a}}$  are integers and  $N - \sum_{\mathbf{a}} d_{\mathbf{a}}$  is even. A nonnegative decomposition additionally requires  $N \geq \sum_{\mathbf{a}} |d_{\mathbf{a}}|$ . The implementation makes these decisions with integer arithmetic. Atomic compatibility concerns the symmetry signature, not a proof of trivial topology. Likewise, a signed-only decomposition does not by itself establish fragile topology. This inversion-only EBR test is distinct from the subsequent native little-group and explicit-segment analysis described below; it does not become a general EBR classifier when that additional analysis is used.

Table S3: Additional native parity checks with four occupied Kramers pairs. The four columns refer to  $(k_x, k_y) = (0, 0), (1/2, 0), (0, 1/2), (1/2, 1/2)$  at  $k_z = 0$ . Coordinates of the chosen inversion center are fractional.

| System  | $\mathbf{c}$      | $n_{\kappa}^{-}$ |   |   |   | $\nu$ | Atomic signature |
|---------|-------------------|------------------|---|---|---|-------|------------------|
| Neon    | $(1/2, 1/2, 1/2)$ | 3                | 3 | 3 | 3 | 0     | Compatible       |
| Stanene | $(0, 0, 0)$       | 1                | 2 | 2 | 2 | 1     | Obstructed       |

These tests retain the basis, cutoff, geometry and SCF meshes of the earlier material examples, replacing the Wilson paths by TRIM sampling. The three-dimensional neon test, with inversion center at the origin, also gives zero strong and weak indices and  $z_4 = 0$ . All seven inputs in the extended `QS/regtest-topology` set passed with two MPI ranks and two OpenMP threads per rank. The stanene parity result was independently reproduced with one rank and one thread. A single-rank, multiple-thread run exposed inconsistent state normalization in the existing local k-point diagonalization path for this build. The new metric checks reject that case. This extension does not resolve or validate that threaded path.

The lightweight CI fixture subsequently reduces the stanene SCF mesh from  $8 \times 8 \times 1$  to  $4 \times 4 \times 1$  and suppresses unused restart files. It still evaluates all four TRIM and reproduces the odd-pair counts  $(1, 2, 2, 2)$  and  $\nu = 1$ . This fixture is distinct from the archived  $8 \times 8 \times 1$  manuscript calculation and does not replace it or establish k-mesh convergence.

The native unit test covers multidimensional-character decomposition, nonorthogonal metrics, degenerate-state gauge changes, broken-symmetry rejection, all inversion-center EBRs and strong,  $z_4 = 2$  and signed-only examples. Independent enumeration of all two-Kramers-pair atomic sums matches the solver for all 81 two-dimensional and 6,561 three-dimensional candidate parity signatures. A deliberately incorrect inversion center is rejected in an end-to-end calculation. TRIM sampling does not prove a gap over the full Brillouin zone, and the SOC subspace still requires its usual convergence checks.

## S5 Native Little-Group and Compatibility Validation

The native algebra constructs unitary projective character tables, Wigner corepresentation restrictions, and endpoint-to-segment branching without a runtime dependency on Spgrep or a pretabulated irrep catalogue. The electronic-structure path uses SPGLIB to detect spatial operations and, when requested, adds their nonmagnetic time-reversal extension. It retains centering translations and tests the metric, band-subspace closure, energy commutators and the complete semilinear multiplication law before accepting band multiplicities.

The all-setting sweep covers all 530 Hall settings of the 230 ordinary space groups. At each setting, it samples  $(0, 0, 0)$ ,  $(1/2, 1/2, 1/2)$ ,  $(1/2, 0, 0)$ ,  $(1/3, 1/3, 0)$ ,  $(0, 0, 1/4)$  and  $(0.173, 0.217, 0.319)$  with scalar and spinful factors. The invariant cell metric is obtained by averaging over the supplied rotations. Both the ordinary group and its grey extension are checked. Generic-site orbits with physical spin multiplicity provide fixed-rank compatible band data. Segments run from Gamma to the other sampled points and from Gamma to its reciprocal image  $(0, 0, 1)$ . This tests directed-image monodromy as well as ordinary endpoint restrictions; it does not sample every possible line or material.

| Checked object                                        | Successful cases |
|-------------------------------------------------------|------------------|
| Ordinary/grey scalar/spinful factor systems           | 12,720           |
| Complete unitary projective character tables          | 6,360            |
| Endpoint tables with an antiunitary stabilizer        | 5,324            |
| Ordinary/grey directed segment comparisons            | 12,720           |
| Segments retaining an antiunitary stabilizer          | 4,462            |
| Gaussian directory assertions, serial / two MPI ranks | 69 / 69          |

Table S4: Native representation validation on 24 September 2026. The counts refer to algebraic cases or numerical assertions, not distinct material calculations.

Analytic cases separately test magnetic half translations, spin quaternion doublets, complex conjugate  $C_3$  pairs, twisted changes of factor gauge, and invalid pairing/doubling counts. Segment cases include displaced-origin PT pairs, endpoint-only time reversal, magnetic-glide Kramers planes, and spinful glide partner switching: a two-band hourglass connection is obstructed, whereas the complete four-band space is compatible. Unitary screw and mirror branching, reciprocal-image monodromy and changed coset representatives are also checked. The complete Hall-setting suite passes in release and with bounds checking and floating-point traps enabled.

The Gaussian directory uses UZH MOLOPT bases for its GPW fixtures and a contracted all-electron basis for GAPW. Ne tests exercise scalar and post-SCF GTH-SOC multiplets at Gamma, a generic PT-symmetric point and nonsymmorphic screw sectors. Scalar He/GPW, He/GAPW and Ne/SOC close a four-segment reciprocal loop; a separately selected single screw branch must report its monodromy obstruction. The generated Wilson-link test also exercises repeated refinement. Serial runs use two OpenMP threads; MPI runs use two ranks with two threads each and the thread-safe BLAS settings stated below. An earlier wider corepresentation integration pass additionally gave 110/110 serial and 111/111 MPI assertions across topology, Kubo, localizer and quadratic directories. These are separate test scopes, not numbers to add into a material benchmark total.

A separate local algebra sweep uses SPGLIB 2.7 magnetic data in the supplied setting for every one of the 1,651 magnetic UNI numbers. At the same six points and two spin choices, all 19,812 factor cases pass; 14,088 contain an antiunitary stabilizer, and 5,724 do not. For 10,154 primitive-cell tables supported by an independent Spgrep 0.7.0 comparison, dimensions, Wigner types and unitary restriction characters agree within  $7.11 \times 10^{-15}$ . Both calculations use the same supplied factor system, so this is not an independent validation of symmetry detection. Moreover, a separate semilinear-product check did not validate every explicit antiunitary matrix produced by that reference calculation. Those matrices are not imported into CP2K; the comparison is limited to types, dimensions and unitary restrictions. CP2K’s actual Gaussian sewing matrices pass their own full product check.

The separate native site-induction test samples fractional positions  $(0, 0, 0)$ ,  $(1/2, 0, 1/4)$  and  $(0.173, 0.217, 0.319)$  at the same six reciprocal points. Ordinary and grey groups, each with scalar and spinful factors, give 38,160 site/k-point cases over all 530 Hall settings. There are 67,824 local-representation columns evaluated and decomposed, and the same number of directed compatibility comparisons. Their atomic band rank equals the orbit multiplicity times the local irrep or corepresentation dimension. This construction preserves conventional-cell translation

sectors rather than silently converting them to a primitive-cell catalogue.

An independent check constructs Cartesian s/p/d orbital traces and, for spinors, multiplies them by the physical  $SU(2)$  trace. After matching the onsite orbital content, its explicitly translated orbit trace agrees with the corresponding induced character sum in 114,480 comparisons. The numerical acceptance tolerance is  $10^{-7}$  for these summed traces and decomposition checks; group and orbit geometry use  $10^{-9}$ . Analytic checks additionally cover inversion-center parities, screw orbits, spin-quaternion doublets, and complex local orbitals related by an antiunitary half translation. These are algorithmic reference tests, not new material calculations or a new production input mode. Additional scalar and spinful checks preserve the characters under passive origin shifts and unimodular changes to a sheared cell, and verify the expected Bloch rephasing under integer changes of operation representatives. The full database sweeps pass in serial, two-rank MPI, and a separate bounds-checked build with floating-point traps.

The native Wyckoff generator separately constructs 3,467 affine site-symmetry families across all 530 Hall settings and determines 5,648 directed specialization relations. External validation uses SPGLIB’s tabulated `Wyckoff.csv` coordinates, not the production fixed-set algorithm, to check dimensions, orbit multiplicities, family equivalence and closure containment. Every family matches bijectively, with largest affine residual  $2.22 \times 10^{-16}$ . The reference table’s version and checksum, comparison script and results are retained in the development record. No tabulated coordinates are read by the CP2K implementation.

For each generated seed the separate spinful induction kernel confirms its orbit and site stabilizer. Combined arbitrary origin shifts, nonorthogonal unimodular cell changes, integer shifts of operation representatives and reordered operations preserve the families and their specialization graph. Analytic tests cover inversion centers, intersecting mirrors, fixed-point-free screw axes, grey groups and an antiunitary half translation. Failure to resolve a generic stabilizer or an arithmetic/work limit returns no partial list. Maximal site groups are identified geometrically; this is not an elementarity test for their induced representations. The complete Wyckoff sweep passes in serial release, two-rank MPI and bounds-checked debug builds.

The subsequent atomic-signature module combines all generated sites and onsite irreps/corepresentations over the six reciprocal points listed above. Ordinary/grey and scalar/spinful variants yield 2,120 matrices, 31,079 provenance-preserving columns and 707,038 integer entries. Seven directed comparisons per column include the supplied point connections and a reciprocal-image loop, totalling 217,553 column/segment checks. Each block reproduces the column’s band rank; antiunitary rows count full corepresentations rather than treating all states as Kramers pairs. Additivity, permuted operation rows, and integer coset rephasing are checked. All sixteen inversion-center signatures and the generic-site signature are recovered analytically.

These full sweeps pass in serial release, two-rank MPI and debug with bounds and floating-point checks. A local optimized-compiler issue in automatic `MATMUL` result resizing was independently reproduced without CP2K and avoided by explicitly dimensioning the test vectors; no physical reference values were adjusted. The compiler reproducer and validation logs are retained separately from material results. No new general-classification input mode, elementarity assertion or complete reciprocal-stratum coverage follows from these matrix tests.

The integer-signature extension factors these same 2,120 matrices using checked 64-bit Smith reduction. Each factorization verifies its full integer certificate. One signed target per matrix is reconstructed, and the nonnegative search recovers every atomic column and a three-column sum per matrix: 33,199 witnesses in total. These deliberately constructed targets test recovery, not a classification of DFT bands.

An independent unit program compares Smith invariants with the gcds of all minors and computes the determinants of both unimodular transforms for 384 matrices with dimensions one through four. The rank-zero through rank-four case counts are 38, 167, 103, 60 and 16. The analytic inversion Fourier criterion supplies a separate exact oracle for every odd-parity-count pattern with two Kramers pairs: among 81 four-TRIM patterns, 41 admit signed and 33 admit nonnegative atomic signatures; among 6,561 eight-TRIM patterns, the counts are 241 and 129.

Thus signed-only cases are tested explicitly. For every  $2 \times 3$  matrix with entries in  $\{0, 1, 2\}$  and every two-component target in  $\{0, \dots, 4\}^2$ , a brute-force enumeration of coefficients zero through four supplies another oracle, covering 18,225 systems. Coefficients beyond four cannot contribute to a target in this range through any nonzero nonnegative column. Zero columns, deficient rank, invalid data, unsafe arithmetic and exhausted node budgets have distinct tests. The latter return an unresolved decision, not a symmetry obstruction.

The expanded matrix tests pass in optimized serial and two-rank MPI runs, and with the integer kernel compiled using bounds checks, floating-point traps and undefined-behavior instrumentation. MPI repeats the native algebra on each rank and does not distribute the catalogue construction. The analytic matrix path also passes with the test's SPGLIB preprocessor option disabled, linked against the existing library; this is not a full no-SPGLIB rebuild.

The Gaussian connection is enabled by `ATOMIC_SIGNATURES`. It compares only physically validated, sampled band subspaces with the generated reference matrix, preserving conventional-cell sectors. Band characters are rephased for integer changes of operation representatives, and the entire product/factor correspondence is checked, including antiunitary conjugation. All 2,120 algebra cases recover the same signature after operation and irrep permutations and such coset shifts. The shifted factors are regenerated by the separate little-group constructor rather than copied from the mapping formula. Inconsistent factors and unmatched operations are rejected. These expanded sweeps pass in optimized serial, two-rank MPI and with the new mapping kernel under bounds and undefined-behavior checks.

In the physical fixtures, scalar Ne at Gamma, Ne/SOC at Gamma and a generic point, centered/sheared He, and complete He/GPW, He/GAPW and Ne/SOC screw branches all have verified signed and nonnegative atomic signature witnesses. The independently obstructed single He screw branch has neither. The refined Wilson fixture repeats the comparison on both its four- and twelve-point sets. An orthorhombic He TRIM fixture sets `ATOMIC_SIGNATURE_MAX_NODES` to zero: signed membership is established, while nonnegative membership remains explicitly unresolved. This separates absence of a witness from proof of absence. The Gaussian-integration driver passed 73 assertions serially and with two MPI ranks, consisting of four native unit programs and 69 Gaussian-directory assertions. Existing physical reference values are unchanged; the new fixture uses the same UZH MOLOPT/GTH family as the other GPW tests. An independent reader then checks the exported records using arbitrary-size integer arithmetic: each of the serial and MPI runs contains 11 matrices and 19 signed/nonnegative witnesses satisfying  $Ax = b$  exactly, with consistent row/column band ranks. This verifies the stored certificates rather than independently classifying the negative cases. The reader and compressed output records accompany `validation/development/gaussian-signature-validation.md`.

The subsequent reciprocal-geometry test enumerates fixed families of  $(-1)^a W^{-T}$  for all 530 Hall settings and their grey extensions. Across these 1,060 geometric groups it retains 18,700 connected fixed-torus closures, 57,357 specialization incidences and 18,730 primitive integer tangent cycles. Full original cosets are restored when checking spinful stabilizers. At each setting, independent direct-transpose tests on reciprocal meshes with denominators 2, 3, 4 and 6 assign every point to exactly one matching family; this gives 333,900 point checks. Joint direct-cell shears, origin shifts, integer coset changes and operation permutations preserve the family and incidence correspondences. Analytic cases separately cover time reversal,  $PT$ , magnetic half translations, screw-invariant lines, centering and orthorhombic intersections.

For each ordinary/grey group, scalar and spinful atomic reference matrices are constructed at the automatically generated seeds and incidence images. Every atomic column obeys each directed restriction and primitive-cycle relation, giving 2,481,572 column/edge comparisons. This explicitly retains reciprocal shifts, nonsymmorphic phases and antiunitary restrictions. The complete sweep passes in serial and on both MPI ranks; the MPI run repeats the algebra, rather than distributing it. The new reciprocal module and its test also pass with bounds checks, floating-point traps and undefined-behavior instrumentation; their dependencies use the existing library. A database-free analytic test build passes against that library, not against a full no-

SPGLIB rebuild. The updated focused driver passes 74/74 serial and 74/74 MPI assertions: five native programs and the unchanged 69 Gaussian assertions. Logs and reproduction details are in `validation/development/reciprocal-strata-validation.md`. These are the geometric-stage results; fixed closures and their incidence/cycle graph are not a complete equivariant open-cell complex or a global connectivity classifier.

### S5.1 Automatic Gaussian symmetry sampling

The subsequent `KPOINTS_SOURCE SYMMETRY` integration generates and refines those paths directly at the frozen SCF Hamiltonian. Four Gaussian fixtures cover scalar He/GPW with UZH MOLOPT/GTH, all-electron He/GAPW with UZH TZVPP functions, Ne/GTH-SOC with its complete scalar AO space in the second variation, and a unimodularly sheared He cell with consistently transformed coordinates. They each yield 15 fixed families, 56 sampled points and 82 directed segments, all compatible. Each has signed and nonnegative atomic-signature witnesses. The focused driver now passes 106/106 assertions serially and with two MPI ranks: five native programs and 101 Gaussian assertions. The 32 added assertions do not alter the previous energy or symmetry references.

An independent file-level replay reconstructs NNKP inputs from the recorded fractional points and reciprocal-image edges. The rectangular NNKP neighbour format needs extra, nonduplicate padding connections; comparisons of segment data use the original directed edges only. Across the four cases, selected eigenvalues differ by at most  $2.1 \times 10^{-13}$  eV and band characters by at most  $2.2 \times 10^{-14}$ . Endpoint-to-line restrictions, the full atomic signature matrix and its integer witnesses agree exactly. The native path creates no `.mmn` file, whereas the replay exercises the existing overlap exporter. This comparison tests the new sampling integration against the established explicit-point path, not an independent electronic-structure implementation.

Increasing the number of points per original segment from three to five gives 138 points and 164 compatible segments for He/GPW. Removing time reversal retains valid unitary restrictions and reports zero antiunitary segments. Negative tests reject a too-small point budget, an invalid subdivision count and a conflicting Wilson-loop request. Selecting one of the two He screw branches fails the isolation check at a generated degeneracy instead of yielding an atomic conclusion. Analytic subdivision tests independently check lifted endpoints, uniform increments and total winding for positive, negative and multiple boundary crossings, including budget failure without mutation. The full 1,060-group geometry sweep is repeated with path subdivision. Records and the replay procedure are retained in `validation/development/automatic-sampling-validation.md`.

### S5.2 Joint compatibility lattice and independent certificates

The following baseline precedes the star-identification extension described in the next subsection; its archived certificates are retained. The next extension assembles all endpoint-to-line restrictions into one integer matrix  $C$ , expressed in the same irrep/corepresentation coordinates as the atomic matrix  $A$ . In particular, a reciprocal-image self-edge subtracts two restrictions in the same point block and retains nonsymmorphic monodromy. An exact Smith kernel  $K$  and integer atomic coordinates  $Y$  satisfy  $CK = 0$  and  $KY = A$ . The Smith form of  $Y$  supplies the free and finite factors of  $\ker_{\mathbb{Z}} C / \text{im}_{\mathbb{Z}} A$ .

The 530-setting ordinary/grey and scalar/spinful sweep verifies every generated atomic column in the joint kernel with zero quotient class. Small-group variants reverse all directed edges and reciprocal shifts, reject a deliberately incompatible atomic column and enforce a tiny memory budget without returning partial data. Separate integer tests include a nonsaturated constraint row, a free-plus-torsion quotient, zero-dimensional kernels and incompatible targets. The four/eight-TRIM inversion matrices with equal-rank constraints yield  $\mathbb{Z}_2$  and  $\mathbb{Z}_2^3 \times \mathbb{Z}_4$ , respectively. For all 81 and 6,561 two-pair parity patterns, zero class is equivalent to the independently tested signed atomic membership.

The full sweep passes in optimized serial, on both MPI ranks and with the new integer/compatibility and reciprocal kernels compiled with bounds, floating-point traps and undefined-behavior checks. All printed graph ranks/factors agree. The instrumented database-free analytic subset also passes against the existing library, not a full no-SPGLIB build. MPI repeats the algebra independently on each rank.

An independent SymPy 1.14.0 reader validates eleven exported Gaussian quotient records from each of the serial and two-rank MPI runs. It checks  $CK = 0$ ,  $KY = A$  and  $CA = 0$ , computes the rational rank of  $C$  and an independent integer Smith form of  $K$  to certify saturation, and factors  $Y$  independently. It also checks unimodularity of the exported quotient row map. Dividing its nonzero transformed rows by their divisors must give a primitive full-row-rank lattice; free rows must vanish. Finally it solves for each compatible target in  $K$  and recomputes the exported residue vector using arbitrary-size integers. Incompatible targets must not have a quotient-class record.

For the automatic He/GPW, He/GAPW and sheared-He fixtures,  $C$  has 109 retained equations and 91 coordinates. Its kernel has rank six, no free quotient factor and one  $\mathbb{Z}_2$  factor. The Ne/SOC fixture has 81 equations, 67 coordinates, kernel rank five and quotient  $\mathbb{Z}_2^2 \times \mathbb{Z}_4$ . All four physical targets have zero class. These are deliberately small atomic examples, not nontrivial topological-material discoveries. The explicit four-point complete screw fixtures have trivial quotients. The selected single screw branch violates compatibility and is assigned no class. By contrast, the sparse explicit He and refined Wilson graphs have free ranks four and 23. Those ranks expose missing constraints or identifications in the supplied graph and are not labelled physical topological invariants. Even a zero free rank does not establish completeness of omitted congruence constraints or BZ-wide isolation.

The official focused and exporter drivers together pass 165/165 assertions in each of SSMP and two-rank MPI. This comprises 129 native and little-group assertions and 36 existing export/k-point assertions; 23 new checks inspect compatibility, kernel rank, free/cyclic factors and actual target classes without changing existing physical reference values. Reproduction commands, the independent reader and compressed certificates accompany `validation/development/compatibility-lattice-validation.md`.

### S5.3 Star transport and published quotient comparison

The subsequent extension includes star-arm identifications in the same integer matrix, using the star-character transport formula derived in the main text. A threefold screw example checks genuinely complex characters, antiunitary conjugation, reciprocal shifts and independent integer changes of destination representatives. A star-only graph reduces nine local coordinates to three independent ones without requiring any line edges; a deliberately mismatched target is rejected. Corrupted factors, nonfinite characters and unsafe phases must not return a permutation. Every retained star link in the complete Hall-setting sweep agrees with independently induced atomic columns; inverse maps and all alternative connecting operations give the same full-corepresentation identification.

An independent development reader compares the resulting ranks and ordered nontrivial Smith factors with Tables I–IV and IX–XII of [Po, Vishwanath and Watanabe \(2017\)](#). It validates all 230 entries in each rank table and uses the SPGLIB Hall-to-space-group metadata solely to associate settings with the reference rows. All 2,120 ordinary/grey and scalar/spinful Hall-setting combinations agree, with no free factors on these automatically generated graphs. These are representation group comparisons, not 2,120 DFT calculations. They do not independently prove completeness of a user-supplied graph or the elementary nature of every generated atomic representation.

The Gaussian certificate reader also checks each exported star’s reciprocal geometry. The explicit He/GPW and He/GAPW screw examples now contain ten equations for nine coordinates, still with compatible rank three and trivial quotient. The Ne/SOC screw has five equations, five

coordinates and rank two. The refined He Wilson graph contains 76 equations and 75 coordinates: its kernel rank falls from 42 to 23 and its free quotient rank from 23 to four. The partial two-point He graph still has four free factors; the single-branch screw obstruction remains incompatible. No energy, character or physical reference value is adjusted to obtain these results. The automatic Gaussian fixtures retain the ranks and finite factors of the baseline.

Reproduction scripts, reference-comparison records and the new certificates are archived separately in `validation/development/star-transport-validation.md`. The star-transport stage passed 170/170 assertions in each of SSMP and two-rank MPI: 134 native/little-group checks and 36 existing export/k-point checks. Five new Gaussian assertions count star links, including the zero-link monodromy control and the six links of the refined Wilson graph. The full star-aware sweep output is identical between optimized SSMP, each MPI rank and the instrumented build. The database-free analytic variant passes as well, using the existing library rather than a full no-SPGLIB build.

These algebra sweeps do not implement magnetic-space-group detection, a magnetic non-collinear ground state, canonical irrep labels, a complete Brillouin-zone connectivity graph or a full elementary-band catalogue. Explicit-segment compatibility is necessary, not sufficient, for a gapped band subspace. The retained summaries and test logs are in `validation/development/`; no production material classification is inferred solely from a successful database sweep.

#### S5.4 Real-space induction and composite witnesses

The subsequent site-specialization extension evaluates finite site-group induction at lifted endpoints of the generated Wyckoff families. All 22,592 relations in the ordinary/grey scalar/spinful 530-setting sweep pass their independently generated band-character identities at eight reciprocal points, giving 180,736 checks. Transitive singleton equivalences and direct-sum witnesses also preserve every tested character and physical band dimension. The optimized SSMP, both MPI ranks and bounds/undefined-behavior instrumented executable produce identical 4,774-line result records. Intermediate-position tests cover inversion and mirror families, cell shear, shifted origin and integer coset changes. A separate analytic build omits the database interface; this uses the existing CP2K library rather than constituting a complete no-SPGLIB build.

The detected exceptional-space-group coverage matches Tables II–IV of [Cano et al. \(2018\)](#) in all 2,120 variants. A second comparison uses the independently archived Bilbao EBR data in `Crystalline.jl`. Every per-setting generator count agrees; counting each space group once gives 3,383 scalar and 2,258 spinful generators without time reversal, and 3,141 scalar and 1,616 spinful generators with time reversal. The archived dataset’s 2,258 differs from the aggregate 2,263 printed in the 2018 article. The comparison retains the dataset’s values and does not adjust a native count to obtain agreement. Neither these totals nor agreement of exceptional-space-group coverage establishes a canonical irrep-by-irrep identification.

For each of 15 Gaussian detail files per build, an independent arbitrary-precision reader verifies  $AE = A$ , the conserved band dimensions, the lifted endpoint geometry, every source stabilizer’s action on the entire displacement, singleton equivalence classes and recursive decomposition provenance. The physical fixtures include GPW, all-electron GAPW, post-SCF GTH SOC, sheared cells, screw sectors, TRIM and refined Wilson sampling. The official serial and two-rank MPI drivers each pass 171/171 assertions: six native unit programs, 129 Gaussian little-group assertions and 36 existing export/k-point assertions. The bounded new unit includes the spinless time-reversal exception at an  $S_4$  site in space group 84.

The reproducible comparison scripts, versioned reference hashes, native logs and Gaussian certificates are retained in `validation/development/site-induction-validation.md`. The reference tables are test oracles, not runtime dependencies or imported CP2K classification tables. These tests prove the returned decompositions and their integration, not a globally gapped material classification, canonical EBR labels or completeness of arbitrary reciprocal sampling.

### S5.5 Affine-reference and site-resolved EBR checks

The next reference layer uses the separately archived space-group generators and parametrized Wyckoff coordinates of Crystalline.jl. The test fixture closes the generators with exact rational arithmetic; the native reference path does not obtain operations from SPGLIB. All 1,731 named affine families match the generated closures, including their whole tangent spaces, dimensions and conventional-cell multiplicities. A second pass simultaneously changes the basis, origin, integer coset representatives and operation order and preserves every named specialization edge. The four ordinary/grey scalar/spinful variants give 6,924 affine correspondences.

After the conventional-cell centering factor is removed, the band dimensions of the 10,398 unreduced maximal-site columns agree with the independent EBR multisets separately at each Wyckoff position. No reference column is selected or discarded using the legacy `Decomposable` field. This test detects incorrect distributions among sites or band dimensions that total counts would miss, but does not assign canonical labels to equal-dimensional irreps. Optimized serial and two-rank MPI output is compared with a separately instrumented reference executable compiled without its SPGLIB database interface. The latter still links the existing CP2K library, and is not a complete no-SPGLIB CP2K build. Numerical guards additionally reject wrong tangents, repeated reference entries, incorrect multiplicities and nonfinite or unbounded coordinates. Reproduction scripts, source hashes and logs are retained in `validation/development/affine-reference-validation.md`.

### S5.6 Complete character correspondence with reference tables

The `irreptables` 3.1.0 package supplies independent single- and double-valued characters at the tabulated reciprocal points of all 230 space groups. The fixture reader uses the data files only, not IrRep’s character-generation implementation. It records a SHA-256 hash for each of the 460 source files. Centering is removed by an explicit primitive basis  $P$ , with  $W' = P^{-1}WP$ ,  $\tau' = P^{-1}\tau$  and  $k_{\text{native}} = -P^T k_{\text{table}}$ . The sign converts the table’s translation-character convention to the native  $\exp(-2\pi i k \cdot T)$  convention; the original character columns and spin lifts are retained. Without this conversion, central-projector audits reject 25 boundary-point tables. With the declared conversion, all 2,700 tables pass the independent audit. This is a convention check, not evidence of an error in IrRep.

Crystallographic coordinates printed to five decimal places are recovered as rational values with denominators at most 48 and changes strictly below  $5.1 \times 10^{-6}$ . Characters are not rationalized or rounded. A Cartesian intertwiner aligns the reference spin matrices with the integer rotations; unitary polar factors are used only to identify this coordinate frame, while comparisons retain the raw reference matrices. Native spin representatives are returned by the same factor-system construction that generated their signs. An independent recomputation after a different floating-point cell normalization is not used to infer those signs.

Every character column is compared, including distinct irreps with the same dimension. All 8,907 columns match; the largest residual is  $1.20921 \times 10^{-5}$ . The native construction tolerance remains  $10^{-9}$ ; the external comparison uses  $2 \times 10^{-4}$  to accommodate the printed reference precision. The 11,607 normalized case and label records are identical in optimized SSMP, separately on both MPI processes, and in a focused GNU Fortran build with bounds checking, floating-point traps and undefined-behavior instrumentation. The instrumented executable links the existing CP2K library; this is not a fully independent no-SPGLIB build or a multi-node scaling test.

The bounded native test covers ordinary and projective  $C_3$ , complex rephasings, label-permuting one-dimensional twists, interleaved antiunitary operations, and paired/doubled Wigner restrictions. An analytic body-centered three-screw boundary case also changes the Cartesian orientation, operation order and cell scale from  $10^{-100}$  to  $10^{100}$ . Inconsistent factor-gauge maps, repeated character columns, NaNs, unbounded values, and orthogonal dimension-preserving mixtures that are not representation characters are rejected. Full onsite EBR naming, arbitrary-cell standardization for named labels, and global connectivity are not implied by these reference correspondences. The

official SSMP and two-rank MPI drivers each pass 173/173 assertions: eight native unit programs, 129 Gaussian little-group checks and 36 existing topology/export/k-point assertions. No physical reference value was changed for this work. Scripts, provenance and numerical logs are retained in `validation/development/character-reference-validation.md`.

### S5.7 Onsite axes, spin phases and reference correspondence

The onsite matcher retains the seed stabilizer’s Cartesian rotations, antiunitary flags, product table and actual spin lifts. An explicitly supplied orthogonal map aligns the reference axes with the native axes. With its spin representative  $V$ , a unitary reference lift transforms as  $VUV^\dagger$ , whereas the matrix part of an antiunitary lift transforms as  $VAV^T$ . Here  $A$  includes  $i\sigma_y$  for physical time reversal. Comparing these matrices with the retained native lifts determines a single phase per operation. The product and factor tables and all character columns must then agree after this declared operation/phase map; phases are not fitted independently to different irreps. The interface returns both the column correspondence and its operation/phase provenance.

For an independent point-group test, the 460 scalar/spinful Gamma tables of `irreptables` 3.1.0 are reused, with their space-group translations removed. Thus the group fixes the chosen site: this is not an enumeration of every actual Wyckoff-site stabilizer in the 230 space groups. Three variants combine cell scales  $10^{-100}$ , 1 and  $10^{100}$ , basis shear, shifted origin, reversed operation order, integer coset changes and Cartesian rotation. Table S5 summarizes the resulting comparison. The  $10^{-9}$  native tolerance is unchanged; the external tolerance is  $2 \times 10^{-4}$  for the finite-precision reference rotations, spin matrices and characters.

| Onsite correspondence check                           | Result                 |
|-------------------------------------------------------|------------------------|
| Gamma reference tables / coordinate variants          | 460/3                  |
| Complete-table matches / matched column records       | 1,380/6,384            |
| Reference-normalized character entries per executable | 88,602                 |
| Largest combined external comparison residual         | $3.14 \times 10^{-5}$  |
| Largest cross-build character difference              | $8.44 \times 10^{-15}$ |
| Largest cross-coordinate character difference         | $9.77 \times 10^{-15}$ |

Table S5: Onsite point-group correspondence validated on 24–25 September 2026. Each MPI process repeats the complete audit; these counts do not measure distributed speedup or material coverage.

In several cubic spinful cases, 52 native-column/label records differ between release and instrumented builds. The corresponding complete characters, indexed by the returned reference labels and operations and divided by the explicit lift phases, agree to the tolerance shown above. Raw internal irrep indices therefore must not serve as cross-build physical identifiers. Both MPI processes retain the release ordering, but correctness does not depend on that coincidence.

The bounded analytic tests distinguish the three complex  $C_3$  characters and their scalar/spinful grey restrictions, test improper axis changes and complex spin-lift rephasings, and reject inconsistent characters, missing antiunitary spin factors, duplicate operations and nonfinite or unbounded input. A nonsymmorphic  $P6_3$  example has two sites and a  $C_3$  onsite subgroup. At  $k_z = 0, 1/4, 1/2$ , operations exchanging sites have zero induced trace; the remaining characters agree with the analytic onsite sum and screw-translation phase. This holds for scalar/spinful ordinary and grey variants after origin, basis, axes and operation-order changes. Separate magnetic tests distinguish an onsite  $C_2\mathcal{T}$  operation, whose spinful square is  $+1$ , from an antiunitary half translation exchanging sites.

The focused instrumented executable checks the symmetry, little-group, corepresentation, character-matching and site-induction modules with bounds checks, floating-point traps and undefined-behavior instrumentation. It omits its SPGLIB database test interface but links the existing release CP2K library; it is neither a fully instrumented Gaussian calcu-

lation nor a complete no-SPGLIB build. The official serial and two-rank MPI directories each pass 179/179 assertions after this extension, without changes to physical reference values. Scripts, input hashes, complete character records and driver logs are retained with `validation/development/onsite-reference-validation.md`. This establishes a convention-aware onsite matching kernel, not automatic standard labels in the Gaussian output, complete actual-site EBR identities or a material topology classification.

## S5.8 Actual Wyckoff sites and independent Bloch induction

The subsequent comparison retains the real fractional space-group translations. The independently archived affine families supply one generic point per family, formed from their origin and tangent parameters  $\sqrt{2}/7$ ,  $\sqrt{3}/7$  and  $\sqrt{5}/7$ . An explicit integer-Hermite construction removes centering and records the primitive basis. All 1,731 conventional families are retained, including nonmaximal sites, rather than selecting a subset by EBR dimension or by the archive’s decomposability field.

The reference reader uses Gamma character tables for 32 symmorphic point-group representatives, separately for scalar and spinful representations. It matches their Cartesian operation sets to each actual site stabilizer by an explicit orthogonal conjugacy and records the chosen axes. For grey groups, independent character conjugation and the spin-aware Frobenius–Schur indicator construct paired or doubled unitary restrictions. These reference axes are not automatically the standard Bilbao onsite-label convention. The native matcher must accept the complete table, including equal-dimensional columns, in both the original and simultaneously rotated, sheared, origin-shifted, reordered and integer-coset-shifted coordinates.

An independent Python calculation then builds the real-space orbit and its representatives. For each unitary operation that fixes an orbit site modulo a translation, it evaluates the onsite character with an explicit physical  $SU(2)$  intertwiner and multiplies it by  $\exp(-2\pi i k \cdot L)$ . Summing these contributions gives the whole-band trace without using CP2K’s cached induction coefficients. The primitive reciprocal points are  $(0, 0, 0)$ ,  $(1/2, 0, 0)$ ,  $(0, 1/2, 0)$ ,  $(0, 0, 1/2)$ ,  $(1/4, 1/4, 1/4)$  and  $(0.173, 0.217, 0.319)$ . Only unitary little-group operations are compared at each point; their complete membership is checked independently. The native results are converted using explicit spin-lift and integer-coset phases, not by fitting a phase to each band representation.

| Actual-site and induction comparison                       | Result                  |
|------------------------------------------------------------|-------------------------|
| Wyckoff families / symmetry variants / coordinate variants | 1,731/4/2               |
| Complete onsite-table matches                              | 13,848                  |
| Onsite columns before coordinate repetition                | 16,957                  |
| Induced tables / character entries per executable          | 83,088/1,384,328        |
| Largest absolute induced reference difference              | $1.4510 \times 10^{-4}$ |
| Largest dimension-scaled reference difference              | $1.05 \times 10^{-5}$   |
| Largest induced cross-build difference                     | $1.87 \times 10^{-14}$  |
| Largest induced cross-coordinate difference                | $2.67 \times 10^{-14}$  |

Table S6: Complete actual-site comparison on 25 September 2026. The external differences reflect the finite-precision reference characters and spin matrices. Native construction uses  $10^{-9}$ ; the external bound is  $2 \times 10^{-4}$  after division by each induced column’s positive band dimension.

The optimized SSMP, both MPI processes and the focused instrumented executable pass all cases. The 189,664 onsite character entries agree across builds within  $8.22 \times 10^{-15}$ ; the larger induced comparison is summarized in Table S6. A deliberate phase reversal at site  $c$  of space group 100 changes one scalar induced character from  $-2i$  to  $+2i$  at the quarter-diagonal point. The test rejects it with absolute error 4 and dimension-scaled error 1. A separate output audit caught and corrected list-directed Fortran parsing of labels such as `2*GM1`; these labels are now read literally, without changing numerical reference characters.

This optional reference sweep adds no external database dependency or input keyword

to production calculations. The standard native unit tests also pass in serial and MPI with two OpenMP threads; the preceding 179/179 Gaussian/export driver results are not counted as new DFT runs for this test-only extension. The focus remains generic representatives and six sampled reciprocal points, not every coordinate in a free family, complete reciprocal-stratum connectivity, magnetic-space-group site enumeration or automatic canonical EBR naming. The archive, source/fixture hashes and replay scripts are recorded in `validation/development/wyckoff-character-validation.md`.

## S5.9 Recoverable onsite provenance in Gaussian output

The atomic-signature output retains the local character columns, their full site-group operation mapping, Cartesian axes, atom-cell shifts and physical spin lifts, together with the lifts of the complete space group. These data are copied from the construction used for the induced columns. No second choice of quaternion signs is made. The onsite product table is written as signed operation indices: its factors are exactly  $\pm 1$  at Gamma, which the writer checks before encoding. This reduces output volume without rounding characters or suppressing validation. Scalar characters do not use the recorded spin-1/2 matrices.

An independent reader reconstructs the orbit sums using only these records, then compares them with the little-group decompositions of the exported atomic columns. For each of the optimized serial and two-rank MPI builds, 16 retained Gaussian test outputs cover 197 site tables, 697 onsite columns and 263 reciprocal queries, giving 31,537 induced character entries. GPW, GAPW, post-SCF SOC, sheared cells, screw phases and automatic reciprocal sampling are included. The largest reconstruction error is  $3.58 \times 10^{-14}$ . All onsite columns also match the external point-group references, with largest combined lift/character residual  $9.56 \times 10^{-6}$ ; the declared axes and column correspondences agree between builds. These are reference-frame identifications, not canonical Bilbao onsite or EBR labels. Deliberate changes to a character, lattice shift, full-group spin entry or product sign are rejected in all 128 negative checks. Both regression drivers pass 179/179 checks; the high-cutoff Ne case uses a 64 MiB OpenMP thread stack to avoid a separately identified local grid-kernel stack overflow. No physical reference value or tolerance is changed. The native snapshot audit additionally passes all 2,120 Hall-setting/symmetry variants. The evidence and replay tools are retained in `validation/development/onsite-export-validation.md`.

## S5.10 Site-resolved named EBR character sequences

The EBR vectors supplied with `irreptables` 3.1.0 are combined with its independently tabulated little-group characters, without using the native induction algorithm to create the reference. The ordinary-group scalar and spinful datasets contain 1,599 site/spin cases and 5,641 named reference columns across all 230 space groups. There are 9,667 reciprocal queries and 297,529 induced character entries per coordinate convention. Every reference column must match the same native onsite column at all supplied points and operations. Candidate sets are retained, and their multiplicities are checked so that duplicate reference columns cannot consume one native column twice.

The earlier little-group fixture lacks some opposite-valley points. Explicit type-I tables supply 62 such points in 40 scalar/spinful tables, including KA/HA and PA/WA. Before use, the adapter verifies the complete operation map, physical spin lifts and integer coset shifts; all shared labelled characters agree within  $3.01 \times 10^{-16}$ . Missing labels are not assigned by guessed complex conjugation. The reference uses the already validated  $\mathbf{k}_{\text{native}} = -P^T \mathbf{k}_{\text{table}}$  convention and retains the printed character precision.

Optimized serial, both processes of a two-rank MPI run and a focused instrumented executable pass the original and combined sheared/rotated/origin-shifted/reordered conventions. They agree on all candidate multiplicities. Within the specified Wyckoff position, 5,263 reference columns have one candidate and 378 have multiple candidates. The largest dimension-scaled character residual is  $1.05 \times 10^{-5}$ , below the unchanged  $2 \times 10^{-4}$  reference tolerance; native construction still

uses  $10^{-9}$ . The MPI processes repeat the audit independently, so these results are not a scaling measurement.

Six deliberately malformed P4 spinful references are rejected by both optimized executables: a boundary character, a shifted site with unchanged Gamma data, a duplicated character column, a zeroed spin lift, a duplicate operation, and unexpected trailing data. The unchanged baseline passes. Instrumentation covers the test and the symmetry, little-group, corepresentation, character-matching and induction kernels; other modules remain release-library code. This is not a fully instrumented Gaussian or MPI calculation. The generated local space contains 5,704 columns at the selected sites; 63 are absent from the supplied EBR references. Their elementarity, grey-group reference matching, canonical onsite axes and complete band connectivity are not certified by this test. Scripts, input and source hashes, full output and replay instructions are retained in `validation/development/ebr-character-validation.md`.

### S5.11 Character functions beyond finite reciprocal samples

Equal characters at selected reciprocal points need not imply equal character functions. We distinguish these cases without replacing the sampled integer signature matrix or its real-space induction witnesses. For a unitary operation with integer direct-space rotation  $W$ , the invariant reciprocal torus is

$$K_W = \{\mathbf{k} \bmod \mathbb{Z}^3 : (W^T - I)\mathbf{k} \in \mathbb{Z}^3\}. \quad (\text{S5})$$

Its character lattice is  $\mathbb{Z}^3/(W - I)\mathbb{Z}^3$ . Indeed, two integer shifts differing by  $(W - I)\mathbf{n}$  give identical Bloch phases on  $K_W$ . For the checked Smith factorization  $U(W - I)V = D$ , each finite induction term  $c_j \exp(-2\pi i \mathbf{k} \cdot \mathbf{L}_j)$  is therefore collected under the key  $U\mathbf{L}_j$ , modulo each nonzero Smith divisor. Rows with zero divisor retain an unrestricted integer key. The coefficients  $c_j$  include the local character, the spin-lift factor and, where required, conjugation by an antiunitary orbit representative. Antiunitary operators themselves have no basis-independent trace and are not assigned character functions.

These quotient characters are linearly independent on the complete compact group  $K_W$ . Equality of the collected coefficients thus tests equality of the finite character functions, including all disconnected torsion components, rather than fitting more  $\mathbf{k}$  points. Integer operations are checked for overflow; complex coefficients are compared at  $10^{-9}$  after scaling by band dimension in this audit. The production diagnostic uses the supplied character tolerance and rejects nontransitive tolerance neighborhoods. It reports local unitary-character classes but does not merge atomic columns, change membership decisions or infer an intertwiner between Bloch bundles.

As an analytic counterexample to Gamma-only matching, equal-parity orbitals at inversion centers  $(0, 0, 0)$  and  $(1/2, 0, 0)$  have the same Gamma characters but different characters on other components of the inversion-fixed torus. The Fourier comparison distinguishes both parity pairs. With only Gamma supplied to the atomic catalogue, all sixteen centered inversion columns and the generic-orbit column remain distinct, also with Kramers pairing.

For the 378 multi-candidate references in the preceding 230-space-group EBR audit, however, every candidate set has equal collected unitary-character coefficients in both tested coordinate conventions. Optimized serial and both MPI processes agree. These ambiguities are therefore not removed by denser sampling; canonical onsite conventions and bundle-level information remain separate requirements. Independent evaluation of the collected Fourier sums against raw Bloch induction checks 213,238 operation/ reciprocal-point queries per optimized executable or MPI process, with a largest dimension-scaled discrepancy below  $4.1 \times 10^{-15}$ . This count includes the bounded native database and analytic controls, not additional material calculations or an MPI scaling measurement. Full logs and replay instructions are retained with the evidence below. A separate all-setting sweep checks 536,410 Fourier evaluations for all 530 Hall settings, ordinary/grey and scalar/spinful groups at three sites, with a largest discrepancy below  $3.8 \times 10^{-16}$ . The updated atomic catalogue constructs 2,120 matrices with 31,079 columns in both optimized builds, while retaining its signed and nonnegative induction witnesses. The 179 relevant native, Gaussian, topology

and k-point regression checks pass in each build. Independent reconstruction from 16 Gaussian detail files per build retains the earlier 31,537 character-entry comparison and rejects 128 deliberately corrupted records across both builds. The focused instrumented run additionally checks 205,614 Fourier evaluations; it includes the induction and checked integer kernels but not the native SPGLIB database sweep or a fully instrumented Gaussian calculation. Data, source hashes and replay instructions are retained in `validation/development/fourier-character-validation.md`.

The Gaussian exports also illustrate why the two diagnostics must remain separate. The counts below refer to generated atomic reference columns, not numbers of occupied bands or Wannier functions. Both optimized builds give the same counts; the largest accepted coefficient residual in these exports is  $1.86 \times 10^{-17}$ .

| Test input                      | Columns | Sampled classes | Full-torus classes |
|---------------------------------|---------|-----------------|--------------------|
| <code>helium.inp</code> (GPW)   | 125     | 18              | 125                |
| <code>helium-gapw.inp</code>    | 11      | 5               | 11                 |
| <code>neon-soc.inp</code>       | 33      | 17              | 29                 |
| <code>neon-spin-lift.inp</code> | 26      | 23              | 23                 |

Here full-torus classes still mean equality of unitary characters only. Their identifiers are local to each constructed catalogue; all original columns and the sampled membership results are retained.

### S5.12 Shared spin lifts in the Gaussian SOC analysis

The Gaussian sewing matrices must use the same  $SU(2)$  representatives as their factor table. This is tested with Ne in the primitive body-centered cubic basis

$$A = 4.2 \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1/2 \end{pmatrix} \text{ \AA}, \quad (\text{S6})$$

with roundoff-sized perturbations of the input components. The calculation uses PBE, the DZVP-MOLOPT-GGA-GTH-q8 basis from `BASIS_MOLOPT_UZH`, post-SCF GTH SOC, all 13 scalar states in the second variation, and the lowest isolated Kramers doublet as the represented subspace. Both the SCF and the property calculation use Gamma. A single 1200 Ry real-space grid resolves the quadrature sufficiently for the unchanged  $10^{-6}$  metric/subspace and  $10^{-6}$  hartree energy-commutator tolerances.

For identical inputs, the independent-lift and shared-lift paths have metric residual  $1.21 \times 10^{-15}$  and subspace residual  $2.90 \times 10^{-7}$ . Their product-law residuals are respectively 2 and  $8.49 \times 10^{-14}$ . The former correctly aborts; the latter identifies one two-dimensional unitary irrep and one undoubled corepresentation, with signed and nonnegative atomic membership. The SCF energy is  $-34.905537447716$  hartree in both runs, differing by less than  $3 \times 10^{-14}$  hartree before rounding. There is no post-hoc fitting of sewing matrices, reference-energy change, or relaxation of the symmetry checks.

Two distinct convergence issues were separated from the lift defect. At 400 Ry, the Gamma calculation has subspace residual  $9.19 \times 10^{-5}$  and fails the closure test before reaching the product test. Also, the shifted  $2 \times 2 \times 2$  Monkhorst–Pack set in this particular primitive basis is preserved by only 8 of the 48 cubic spatial operations; increasing its real-space cutoff does not restore the missing integration-set symmetry. These diagnostic failures are retained, not converted to successful references. The shared-lift native test additionally applies all 48 cubic operations and their 48 time-reversed partners to complex frames, including cell scales from  $10^{-100}$  to  $10^{100}$ . With this additional Gaussian test the official SSMP and two-rank MPI drivers each pass 179/179 assertions. The focused spin/factor unit tests also pass with bounds checking, floating-point traps and undefined-behavior instrumentation; this is not a fully instrumented Gaussian or MPI build. The reproduction record is `validation/development/gaussian-spin-lift-validation.md`.

## S6 Bloch Kubo and Property-Symmetry Validation

The retained development report is `validation/development/bloch-response-summary.md`. It records distinct finite-cutoff regression fixtures and the higher-cutoff primitive/supercell comparison. For Ne with post-SCF GTH SOC and PBE at 2000 Ry, the  $3 \times 1 \times 2$  primitive mesh gives  $-34.923548219306689$  hartree, compared with  $-209.541289276241002$  hartree for six cells sampled explicitly at Gamma. Both print  $6.21105320 \times 10^{-9}$  S. At the inexpensive 200 Ry PADE regression settings, the respective conductivities are  $6.42389197 \times 10^{-9}$  and  $6.35401974 \times 10^{-9}$  S. Those distinct references are retained: the mismatch is not removed by changing a reference value.

Ne/SOC on a  $2^3$  property mesh uses four representatives. Bi<sub>2</sub>/SOC on  $3^3$  uses six; its printed nine-component tensor differs from full-grid by at most  $2.71 \times 10^{-15}$  relative to its largest entry. Primitive diamond/SOC on  $2^3$  uses three representatives with either K290 or SPGLIB. Its tensor difference is  $2.21 \times 10^{-11}$  in the same normalization. Metric, Hamiltonian and current reconstruction residuals are approximately  $2.1 \times 10^{-13}$ ,  $6 \times 10^{-11}$  and  $1.3 \times 10^{-11}$ , respectively. These are operator-covariance fixtures, not converged material transport predictions.

The final recorded directory drivers passed 132 serial and 162 MPI assertions. The MPI layout was four ranks with two OpenMP threads, and the serial build also used two OpenMP threads. Coverage includes moments, k-point/Wannier export, topology, Kubo, localizer, and quadratic directories; the MPI build additionally tests MUMPS and Tacho. Mathematical unit tests cover the constituent operator identities separately.

The Apple ARM runtime required one BLAS thread, a consistent thread-safe OpenBLAS preload, and an OpenMP stack of 128 MiB. Without these settings, failures occurred before the property code. An implicit-Gamma high-cutoff conversion also exposed a slow Miller-index lookup; the completed comparison uses an explicit Gamma input. These runtime qualifications must accompany the results, rather than be presented as production scalability.

## S7 Localizer and Quadratic Validation Records

The transferred developer notes in `validation/development/` document finite and periodic formulations, dense and sparse factorization, and their limitations. They are implementation records, not independent material benchmarks. The matrix tests include nontrivial and trivial Chern models, spin-mixed Kramers-pair models, origin and energy shifts, nonorthogonal congruences, invalid metrics, and gap closings.

Dense LDL tests cover both  $1 \times 1$  and  $2 \times 2$  blocks. Pfaffian tests independently cover odd permutations and recursive small-matrix reference polynomials. Sparse gap brackets are compared with generalized dense spectra, not with pivot sizes. Realification duplicates the spectrum and is accounted for in every inertia comparison.

The quantitative records below are retained calculations from 24 September 2026, not new material runs performed during manuscript editing. Their raw logs, molecular inputs, state coefficients and model-test source snapshot are archived in `validation/development/solver-test-records.tar.gz`. The accompanying `summarize_solver_tests.py` independently re-parses these records, checks the reported gap brackets and recomputes the molecular metric-subspace comparisons. It writes `solver-test-summary.json`, including SHA-256 checksums for all archived members. Model indices are asserted by the original unit driver; its performance lines do not print a separate index.

### S7.1 Sparse inertia: accuracy and indicative cost

The open square Qi–Wu–Zhang fixture has two orbitals per site, onsite block  $-\sigma_z$ , and positive-direction hopping blocks  $(\sigma_z - i\sigma_x)/2$  and  $(\sigma_z - i\sigma_y)/2$ . The query is at the lattice center and zero energy, with  $\kappa = 0.25$  in model energy per lattice spacing. All nine configurations in Table S7 give index  $-1$ . The dense generalized gap lies within the sparse bracket to the driver’s  $10^{-9}$

comparison tolerance, both before and after the tested nonorthogonal congruence. The printed gap is 0.81017072 for side length 10 and 0.81017480 for lengths 20 and 30. These are model energy units, not first-principles hartree energies.

| Side | Real order | Ranks | Dense (s) | MUMPS (s) | Fill ratio | Memory (MB) |
|------|------------|-------|-----------|-----------|------------|-------------|
| 10   | 800        | 1     | 0.03321   | 0.05853   | 15.0       | 1           |
| 10   | 800        | 2     | 0.03326   | 0.05922   | 15.2       | 2           |
| 10   | 800        | 4     | 0.03427   | 0.05682   | 15.1       | 4           |
| 20   | 3200       | 1     | 1.89561   | 0.51263   | 21.9       | 6           |
| 20   | 3200       | 2     | 1.88879   | 0.48799   | 22.3       | 9           |
| 20   | 3200       | 4     | 1.89937   | 0.33720   | 22.5       | 17          |
| 30   | 7200       | 1     | 22.94493  | 2.40084   | 36.0       | 19          |
| 30   | 7200       | 2     | 23.11979  | 2.20658   | 37.2       | 32          |
| 30   | 7200       | 4     | 23.23512  | 1.36989   | 36.4       | 45          |

Table S7: Single final runs of the open-model localizer kernel. Real order is twice the complex localizer order. Dense means source-rank generalized diagonalization plus the LDL cross-check; MUMPS includes symbolic analysis, metric checks and all gap shifts. Fixture and coordinate-list construction are excluded. Fill ratio is the maximum factor-entry count divided by input entries; memory is MUMPS’s process-summed solver allocation, not application RSS.

The recorded host is an Apple M4 Max with 14 cores and 36 GiB, using GNU Fortran 16.1, OpenMPI 5.0.9, OpenBLAS 0.3.33 and MUMPS 5.9.1 with PORD ordering. Both BLAS and OpenMP use one thread in these kernel measurements. The larger four-rank run recovers from insufficient factor workspace by increasing the workspace estimate, without changing matrix entries or pivot tolerances. The increasing fill ratio and summed memory make clear that sparse does not imply linear storage or ideal MPI speedup. The test retains dense oracle matrices, so its process RSS would not measure the production sparse path alone. No error bars or statistical performance claim follow from these single final runs.

The near-zero cases check a different property: a small but resolved gap must be distinguished from an unresolved or singular localizer. With the index acceptance threshold  $10^{-8}$ , a dense gap of  $10^{-6}$  is enclosed by  $[9.9994941 \times 10^{-7}, 1.0000076 \times 10^{-6}]$  and accepted. A gap of  $10^{-10}$  is bracketed by  $[5.8207661 \times 10^{-11}, 1.1641532 \times 10^{-10}]$  but receives no index; the exactly singular case also receives no index. For  $H = \begin{pmatrix} 1 & 20 \\ 20 & 399 \end{pmatrix}$ ,  $S = 2I$ , and  $L = \text{diag}(H, -H)$ , the generalized gap is  $1.2499922 \times 10^{-3}$ , before and after a nonorthogonal basis change. The sparse interval is  $[1.2499921, 1.2499922] \times 10^{-3}$  at printed precision. This example tests the spectral pencil, not an estimate from LDL pivot magnitudes. All intervals are finite-precision numerical brackets, not interval-arithmetic certificates.

## S7.2 Sparse Pfaffian signs with a spin-mixed model

Time-reversed copies of the side-10 model are coupled by an onsite spin-mixing amplitude  $0.1 + 0.05i$  with the antisymmetric orbital structure required by physical time reversal. The complex localizer has order 800. The nontrivial case, a mass-shifted trivial case, and a complex nonorthogonal transformation of the nontrivial case give the results in Table S8. Both dense and sparse paths use the same trivial-reference orientation. MUMPS checks the metric and gap independently of the Tacho skew factorization.

Separate polynomial/permutation tests cover 338 dense/sparse Pfaffian comparisons of orders 2–24, scales from  $10^{-120}$  to  $10^{120}$ , disconnected blocks, and odd simultaneous row/column permutations. Deliberately singular cases emit zero-pivot diagnostics and are checked as rejected outcomes, not counted as successful topological classifications. Broken time reversal and invalid metrics are likewise rejected. The factorization itself remains on one rank; these tests do not demonstrate MPI-distributed Pfaffian factors or large SOC-material scaling.

| Case                          | $\nu$ | Gap (model units) | Solve-probe residual   |
|-------------------------------|-------|-------------------|------------------------|
| Spin-mixed nontrivial pair    | 1     | 0.79004778        | $2.28 \times 10^{-14}$ |
| Trivial pair                  | 0     | 1.1109451         | $2.54 \times 10^{-16}$ |
| Nonorthogonal nontrivial pair | 1     | 0.79004778        | $3.88 \times 10^{-15}$ |

Table S8: Dense/MUMPS/Tacho model comparison. Gap values agree at printed precision; the solve probes diagnose the sparse skew factors but are not rigorous global backward-error bounds.

### S7.3 Periodic AO localizer: mesh versus supercell

Ne is placed at  $(3, 3, 3)$  Å in a  $6 \times 6 \times 6$  Å cell periodic in XZ. The PBE/GPW calculation uses DZVP-MOLOPT-GGA-GTH-q8 from `BASIS_MOLOPT_UZH`, GTH-PBE-q8 SOC parameters, a single 2000 Ry grid, and an SCF tolerance of  $10^{-10}$ . A full Gamma-centered  $3 \times 1 \times 2$  SCF/property mesh is compared with a six-cell Gamma supercell and property mesh  $1 \times 1 \times 1$ . The query is  $(2.25, 3, 4.125)$  Å,  $E = -0.1$  hartree and  $\eta = 0.5$  hartree. Both spinor localizers have order 312 and give  $\nu = 0$ .

The generalized gap is 0.380994261391133 hartree for the primitive mesh and 0.380994261819095 hartree for the supercell, a difference of  $4.28 \times 10^{-10}$  hartree. The corresponding total energies are  $-34.923548219306682$  and  $-209.541289276241173$  hartree; normalizing the latter to one primitive cell leaves a difference of  $6.60 \times 10^{-9}$  hartree. This checks the full-torus Fourier construction with separately converged SCF potentials. It is neither an irreducible-property-mesh test nor a nontrivial topological material result. The high cutoff reduces grid error for this comparison; it does not establish a complete basis, size or  $\eta$  convergence study.

### S7.4 Molecular quadratic states and degenerate subspaces

Bi<sub>2</sub> is aligned along x with bond length 2.8 Å in an isolated 8 Å box. The PBE calculation uses TZVP-MOLOPT-GGA-GTH-q5, GTH SOC, a 150 Ry cutoff, and SCF tolerance  $10^{-9}$ . At the molecular center,  $E = -0.1$  hartree and  $\kappa = 0.1$  hartree/bohr, four states are requested in a 68-dimensional spinor AO space. The iterative trial space is bounded at 32 vectors, with a  $10^{-9}$  eigenpair-residual threshold. The GPW and GAPW pseudopotential inputs each compare dense and iterative solvers on four MPI ranks and two OpenMP threads.

The two Kramers pairs have quadratic gaps 0.319693242355779 and 0.320631445276286 hartree (the latter rounded across the pair). In both methods, the largest dense/iterative gap difference is  $6.0 \times 10^{-15}$  hartree. The iterative solve takes 145 Ritz iterations and its largest reported eigenpair residual is  $8.22 \times 10^{-10}$  hartree<sup>2</sup>, compared with  $2.82 \times 10^{-15}$  hartree<sup>2</sup> for the dense solution. Thus agreement of the eigenvalues alone would overstate the accuracy of the iterative eigenvectors.

Reanalysis of the stored coefficients verifies  $C^\dagger SC = I$  to  $1.1 \times 10^{-14}$  and all four singular values of  $C_{\text{dense}}^\dagger SC_{\text{iterative}}$  to within  $10^{-10}$  of unity. This compares the complete subspaces, not arbitrary individual phases or rotations within a Kramers pair. The reported decomposition  $\lambda = \delta E^2 + \kappa^2 \sum_j \delta x_j^2$  is consistent to  $2.9 \times 10^{-15}$  hartree<sup>2</sup>. These residuals refer to projected AO positions, not independently integrated continuum position variances. The small finite-cutoff calculation validates the solver and SOC operator integration; it is not a basis-converged molecular localization prediction.

### S7.5 Periodic quadratic coordinates and solver comparisons

The additional checks on 25 September 2026 use the paired trigonometric AO operators defined in the main text. The raw inputs, output logs, quadratic-state coefficients, source snapshot, and comparison script are retained in `validation/development/periodic-quadratic-records.tar.gz`. The accompanying `periodic-quadratic-summary.json` records the individual values and comparison bounds. The archive is small because it excludes restart wavefunctions and regenerable builds.

Unless specified otherwise, one He or Ne atom is placed at (2.25, 2.25, 2.25) Å in a cubic 4.5 Å cell. The scalar SCF uses LDA/PADE, a single 300 Ry grid, relative cutoff 40 Ry, and SCF tolerance  $10^{-10}$ . Basis functions are DZVP-MOLOPT-GGA-GTH-q2 for He and q8 for Ne from BASIS\_MOLOPT\_UZH; the GTH-PBE potentials, including Ne SOC parameters, come from GTH\_SOC\_POTENTIALS. These deliberately fixed small integration fixtures are not basis-converged predictions or recommended material settings. Both scalar and property meshes are Gamma-centered. The query is (1.875, 2.0, 2.625) Å,  $E = -0.1$  hartree, and  $\kappa = 0.1$  hartree/bohr. Two low quadratic states are requested, with iterative subspace limit 24, iteration limit 500, and eigenpair tolerance  $10^{-10}$  hartree<sup>2</sup> at these eigenvalues.

| Fixture     | Periodicity | Mesh                  | Lowest gap<br>(hartree) | Maximum gap difference<br>(hartree) |
|-------------|-------------|-----------------------|-------------------------|-------------------------------------|
| He, GPW     | Y           | $1 \times 2 \times 1$ | 0.358631854242405       | $1.0 \times 10^{-15}$               |
| He, GAPW    | XY          | $2 \times 1 \times 1$ | 0.351912947825943       | $< 10^{-15}$                        |
| He, GPW     | XYZ         | $2 \times 2 \times 1$ | 0.385841580050524       | $< 10^{-15}$                        |
| Ne, GPW/SOC | XZ          | $2 \times 1 \times 1$ | 0.406734402392864       | $5.0 \times 10^{-15}$               |

Table S9: Periodic quadratic dense/iterative comparisons. The gap difference is the maximum over the two requested states, comparing serial LAPACK with two-rank MPI/DBCSR/MUMPS. Each process uses two OpenMP threads and one OpenBLAS thread. Values use the recorded printed precision; vanishing differences do not imply exact arithmetic.

Including the reduced-mesh, wrapped-query and implicit-Gamma supercell variants gives 14 compared gaps, all within  $5.0 \times 10^{-15}$  hartree. The largest dense/iterative difference in the energy residual is  $1.94 \times 10^{-13}$  hartree. The iterative Ne/SOC eigenpair residual reaches  $6.67 \times 10^{-11}$  hartree<sup>2</sup>; it is reported separately from eigenvalue agreement and is not a bound on an individual eigenvector within a degenerate pair. No claim of gauge-matched periodic AO coefficients is inferred from these gap comparisons. Re-parsing all successful cases, including the additional cell and mesh variants, recovers the energy/chord decomposition of the squared gap within  $4.49 \times 10^{-15}$  hartree<sup>2</sup>.

For He, switching full-mesh SCF to atomic symmetry reduction or shifting the query by torus vectors to (10.875, -7.0, 7.125) Å changes the gaps by at most  $1.1 \times 10^{-15}$  hartree. The analysis still assembles the full torus; only the preceding SCF is symmetry reduced. Equivalent MONKHORST-PACK, unshifted MACDONALD and explicitly listed GENERAL SCF meshes also agree within  $1.1 \times 10^{-15}$  hartree.

An independently converged  $2 \times 2 \times 1$  He supercell, with property mesh  $1 \times 1 \times 1$ , gives gaps 0.385841580972906 and 0.391994927003925 hartree through the implicit-Gamma path. The largest difference from the primitive mesh is  $9.30 \times 10^{-10}$  hartree. The primitive SCF energy is -2.832198277992086 hartree; the supercell energy -11.328793115636264 hartree differs by  $9.17 \times 10^{-10}$  hartree after division by four. An explicit complex Gamma KPOINTS calculation agrees with the implicit-Gamma gaps within  $2.2 \times 10^{-12}$  hartree. The latter comparison checks the two input paths, not exact equality of separately converged SCF potentials or convergence in basis, mesh and cutoff.

Axis-permuted He chains cover X, Y and Z; GAPW slab variants cover XY and YZ, with Ne/SOC exercising XZ. A skew cell with rows (4.5, 0, 0), (0.6, 4.4, 0) and (0.3, 0.2, 4.6) Å is compared with the rigidly rotated cell, atom and query using

$$R = \begin{pmatrix} 0.6 & -0.8 & 0 \\ 0.8 & 0.6 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The resulting gaps 0.392476754147977 and 0.412631258413744 hartree are preserved within  $1.1 \times 10^{-15}$  hartree. This checks input-frame handling, not invariance under changing the primitive

basis of the reciprocal-coordinate localization metric. Separate invalid inputs are rejected for a mesh along an open direction, a mesh supplied with the finite formulation, an oversized AO torus, and inconsistent CELL/POISSON periodicity.

The low-level seven-observable tests independently check exact diagonal spectra, complex hopping, nonsingular AO basis changes, paired sine/cosine origin rotations and query wrapping. These algebraic tests supplement, rather than replace, the Gaussian operator comparisons. They also pass with bounds checking, undefined-behavior instrumentation and floating-point traps in the numerical kernel. The complete affected quadratic/localizer regression directories pass 52/52 serial and 87/87 two-rank MPI assertions, retaining the existing finite-system references. Neither the small torus tests nor the local two-rank runs establish large-system or multi-node scaling.

## S7.6 Complex translation queries and distributed adjoints

These additional checks concern the low-level quadratic solver, not Gaussian band unfolding. The complete source fixtures and raw logs are retained in `validation/development/translation-quadratic-records.tar.gz`; its checksummed manifest and the accompanying summary script preserve the distinction between spectra, operator errors and eigenpair residuals. All model numbers use a lattice spacing of one and an energy scale  $t = 1$ , not atomic units.

The first fixture is a periodic 18-site nearest-neighbor chain. With indices modulo 18,  $T_{i,i+1} = 1$  and  $H = -t(T + T^\dagger)$ . At  $k = 2\pi(5/18)$ ,  $E/t = -2\cos k + 0.031$  and  $\kappa_T/t = 1.3$ , the lowest quadratic eigenvalue is  $(0.031t)^2$ . The numerical state has  $\langle T \rangle = e^{ik}$  and a vanishing translation residual within  $10^{-12}$ . This checks the translation phase itself, not only its real part.

A second periodic fixture has six three-site cells, hopping amplitudes  $-0.8t, -0.8t, -1.2t$  and  $\kappa_T/t = 0.7$ . For each supercell phase  $q = 2\pi m/6$ ,  $m = 0, \dots, 5$ , an independent three-dimensional Bloch block sets  $T_{12} = T_{23} = 1$ ,  $T_{31} = e^{iq}$  and  $H_{12} = H_{23} = -0.8t$ ,  $H_{31} = -1.2te^{iq}$ , with Hermitian completion of  $H$ . The five energies  $E/t = -2.2, -1, 0, 0.7, 2$  and 13 momenta  $k = -\pi + 2\pi j/13$ ,  $j = 0, \dots, 12$ , define 65 queries. The complete 18-dimensional spectrum at every query is compared with the sorted union of all six Bloch-block spectra. All 1,170 eigenvalues agree within  $1.60 \times 10^{-14}t^2$ . This comparison checks the cell-folding construction rather than selected minima of an unfolding plot.

For the open-chain test, the wraparound translation is removed, the trimerized hoppings are retained only between neighboring sites, and site 4 has onsite energy  $0.17t$ . The query is  $E/t = 0.33$ ,  $z = e^{0.73i}$  and  $\kappa_T/t = 1.3$ . A separately assembled normal-product reference uses LU metric solves and a generalized Hermitian eigensolver. Its full spectrum agrees with the quadratic solver within  $3.56 \times 10^{-15}t^2$ . Writing  $T = C + iD$  and instead adding the two Hermitian residual squares drops  $i[C, D]$ ; the largest ordered-eigenvalue discrepancy is  $8.604947 \times 10^{-2}t^2$ . This is a deliberate negative control, not numerical noise or an adjustable reference value.

A nonsingular complex upper-bidiagonal basis change sets  $F_{\ell\ell} = 1 + 0.02\ell$  for  $\ell < 18$ ,  $F_{18,18} = 1$ , and  $F_{\ell,\ell+1} = 0.21 - 0.12i$ . In the transformed problem,  $S = F^\dagger F$  and  $A' = F^\dagger A F$ . Spectra change by at most  $5.33 \times 10^{-15}t^2$  and complex expectations, expressed in model units, by  $2.25 \times 10^{-14}$ . Complex shifts of each operator and its query leave the residual spectrum invariant. Restarted iteration with ten trial vectors obtains the two lowest states; their projector agrees with the dense reference within  $10^{-8}$  in maximum element norm. The test compares the subspace, not arbitrary individual eigenvector phases.

The distributed fixture uses six AO blocks of size three, a complex nonorthogonal metric and a nonnormal complex translation-like operator. Serial and one-, two- and four-rank MPI runs cover both row- and column-oriented process grids. Forward and adjoint products against dense references differ by at most  $4.45 \times 10^{-16}$  in the scaled model units; the largest metric-solve residual is  $4.58 \times 10^{-16}$ . Dense-factor and MUMPS-factor iterations agree with the two reference eigenvalues within  $8.78 \times 10^{-15}t^2$ , while their eigenpair residuals reach  $9.84 \times 10^{-11}t^2$ . Supplying the forward operator in place of its adjoint is deliberately tested and rejected. These tiny MPI fixtures check communication and conjugation, not speedup or memory scaling.

Focused instrumented tests also pass bounds checking, undefined-behavior checks and floating-point traps in the numerical kernel. Traps are temporarily disabled only around external LAPACK calls. Following the complex-query extension, the affected quadratic/localizer directories pass 53/53 serial and 88/88 two-rank MPI assertions, with two OpenMP threads per process and one BLAS thread. Earlier finite and periodic Gaussian regression references remain unchanged.

The model tests above do not, by themselves, validate translated Gaussian integrals or the continuum projection complement. The additional finite-Gaussian checks are reported separately below.

## S7.7 Gaussian translations and continuum projection leakage

The FORMULATION TRANSLATION adapter evaluates  $T_{\mu\nu}(\mathbf{a}) = \langle \phi_\mu | U(\mathbf{a}) | \phi_\nu \rangle$  with  $U(\mathbf{a})\psi(\mathbf{r}) = \psi(\mathbf{r} + \mathbf{a})$ . To test this independently of its pair assembly, a second CP2K calculation adds ghost atoms with the same bases at  $\mathbf{R}_B - \mathbf{a}$ . The cross block of this enlarged system’s ordinary overlap matrix supplies a separate  $T$  reference. Printed  $H$  and  $S$ , this ghost-overlap block, and the exported complex quadratic states determine an independent generalized normal Gram and all observable checks. The reference uses NumPy metric solves and Hermitian diagonalization, not the new adapter. It shares CP2K’s existing Gaussian-integral routines, so a separate analytic primitive-Gaussian case also checks their convention.

The four fixtures use a 7 Å isolated box, PBE, a 150 Ry cutoff, 30 Ry relative cutoff,  $\epsilon_{\text{SCF}} = 10^{-10}$ , and  $\epsilon_{\text{default}} = 10^{-12}$ . Single atoms are at (3, 3, 3) Å; H<sub>2</sub> has its second atom at (3.74, 3, 3) Å. The query is  $E = -0.1$  hartree,  $\kappa_T = 0.3$  hartree,  $\mathbf{a} = (0.63, -0.37, 0.21)$  Å and  $\mathbf{k} = (0.31, -0.22, 0.18)$  bohr<sup>-1</sup>, with no extra  $2\pi$  in  $e^{i\mathbf{k}\cdot\mathbf{a}}$ . The smallest fixture has one normalized  $s$  Gaussian with exponent  $\alpha = 0.7$  bohr<sup>-2</sup>, for which  $T = \exp(-\alpha|\mathbf{a}|^2/2)$ . The remaining systems use TZVP-MOLOPT-GGA-GTH bases and GTH-PBE potentials from the UZH files:  $q = 2$  for He,  $q = 1$  for H, and  $q = 8$  for Ne. Ne uses GAPW, the others GPW. One quadratic state is requested for the single Gaussian and three for the other cases. Iteration uses up to 14 trial vectors, 2,000 iterations and a  $10^{-10}$  relative eigenpair-residual tolerance.

Table S10: Finite Gaussian translation verification. The two error columns are maximum absolute discrepancies in quadratic eigenvalues against the independently assembled ghost-overlap reference, in hartree<sup>2</sup>.  $\ell$  is the dimensionless norm outside the AO span, not a solver error. Dense runs are serial; iterative runs use two MPI ranks and reusable MUMPS metric factors.

| Fixture               | AOs | Dense error           | Iterative error       | Max. $\ell$ |
|-----------------------|-----|-----------------------|-----------------------|-------------|
| He, one Gaussian      | 1   | $2.3 \times 10^{-16}$ | $2.3 \times 10^{-16}$ | 0.87416     |
| He, TZVP              | 6   | $6.1 \times 10^{-12}$ | $5.0 \times 10^{-12}$ | 0.56932     |
| H <sub>2</sub> , TZVP | 12  | $2.9 \times 10^{-13}$ | $3.0 \times 10^{-13}$ | 0.56991     |
| Ne, TZVP/GAPW         | 17  | $9.0 \times 10^{-13}$ | $9.1 \times 10^{-13}$ | 0.47708     |

Across these eight base calculations the largest independently evaluated generalized-eigenpair residual is  $1.13 \times 10^{-11}$  hartree<sup>2</sup>. The maximum element error in  $C^\dagger SC - I$  is  $3.2 \times 10^{-15}$ , in complex translation expectations  $1.6 \times 10^{-15}$ , and in the squared full/projected/leakage decomposition  $1.1 \times 10^{-14}$ . Repeating each calculation with  $-\mathbf{k}$  changes the gaps by at most  $2.0 \times 10^{-14}$  hartree. Zero-translation controls have full residuals at or below  $2.11 \times 10^{-8}$ , corresponding to cancellation at approximately machine precision in the squared norm. Converged Ritz vectors are explicitly renormalized using a fresh  $S$  application, and their true operator residuals are rechecked; accumulated cached metric images are not used to certify the final norms.

A compressed-unitary model independently verifies the continuum complement. Compressing a three-state cyclic translation to two basis functions gives  $T_{12} = 1$  with all other entries zero. With  $H = \text{diag}(0.2, -0.4)$ ,  $E = 0.1$ ,  $z = e^{0.73i}$  and  $\kappa_T = 0.7$  in model units, the two quadratic eigenvalues agree with the analytic  $2 \times 2$  formula within  $10^{-15}$ . The discarded norm is exactly

$\ell^2 = |c_1|^2$ . Complex nonorthogonal basis changes preserve it. An expanding matrix  $T = 2I$ , which cannot be a compression of a unitary, is rejected rather than assigned negative leakage.

Additional dense/four-rank iterative comparisons cover all-electron  $\text{H}_2/\text{GAPW}$  with SVP-MOLOPT-GGA-ae and post-SCF GTH-SOC Ne with DZVP-MOLOPT-GGA-GTH-q8. Gaps differ by at most  $7.5 \times 10^{-14}$  and  $1.62 \times 10^{-10}$  hartree, respectively, including their independently converged SCF potentials. Iterative eigenpair residuals stay below  $9.2 \times 10^{-11}$  hartree<sup>2</sup>; opposite momenta give the same gap spectra within  $6.0 \times 10^{-15}$  hartree. These SOC tests use a restricted SCF followed by GTH SOC, not self-consistent noncollinear DFT. Five negative inputs check missing vectors, incompatible position/scale keywords, the wrong formulation and an inappropriate analysis mesh.

The new GPW/GAPW regression fixtures use two translations and two wave-vector queries; dense and iterative paths share the same gap and leakage references. No earlier regression reference is changed. Focused numerical-kernel tests pass bounds checks, undefined-behavior checks and floating-point traps. After formatting and rebuilding, the quadratic/localizer regression subset passes 57/57 serial and 96/96 two-rank MPI assertions, including the available MUMPS and Tacho paths. Source snapshots, inputs, AO matrices, complex states, logs and checksum manifest are retained in `validation/development/gaussian-translation-records.tar.gz`. The companion replay scripts recompute the above comparisons from the raw archive. These tests establish finite AO translation analysis, not periodic material unfolding, boundary suppression, basis convergence or multi-node scaling.

## S7.8 Periodized translations and independent Gamma supercells

The periodic translation construction uses the full Born–von–Karman AO space. For cell-copy indices  $i, j$  and torus lattice matrix  $h_N = h \text{diag}(N_1, N_2, N_3)$ , the block element is the sum of ordinary Gaussian overlaps with ket displacement  $\mathbf{R}_B + \mathbf{R}_j - \mathbf{R}_A - \mathbf{R}_i + h_N \mathbf{n} - \mathbf{a}$ . Only periodic axes are summed. Reciprocal-plane distances bound all potentially overlapping images, also in a skew cell. The full continuum normal Gram and leakage decomposition from the previous subsection remain applicable. The Hamiltonian and overlap use the same frozen-potential full-cell image lifting as the periodic localizer; no irreducible-SCF weights enter.

The independent reference again obtains  $T$  from ordinary overlap matrices of a separate shifted ghost-basis calculation. Here it uses a three-cell Gamma supercell, whereas the primitive calculation has a Gamma-centered  $3 \times 1 \times 1$  mesh. Both SCFs are converged independently. The original Gamma Hamiltonian and overlap, not the ghost-system Hamiltonian, enter the NumPy generalized normal Gram. Additional overlap images printed during the Gamma-to-k-point setup must not be mistaken for the original Gamma overlap matrix.

He occupies the fractional position  $(1/2, 1/2, 1/2)$  of the primitive cell. The cubic cell has side 4.5 Å; the skew cell has column vectors (4.5, 0, 0), (0.7, 4.8, 0) and (0.3, 0.4, 5.1) Å. TZVP-MOLOPT-GGA-GTH-q2 and GTH-PBE-q2 are taken from the UZH files. These tests use PADE LDA, a 300 Ry cutoff, one grid, 40 Ry relative cutoff,  $\epsilon_{\text{SCF}} = 10^{-11}$  and  $\epsilon_{\text{default}} = 10^{-12}$ . The query parameters are  $E = -0.1$  hartree,  $\kappa_T = 0.3$  hartree,  $\mathbf{a} = (0.7, -0.3, 0.4)$  Å and  $\mathbf{k} = (0.21, 0.10, -0.31)$  bohr<sup>-1</sup>. The torus has 18 scalar AOs. Three quadratic eigenpairs are requested; iterative runs allow 14 trial vectors, 2,000 steps and a  $10^{-10}$  eigenpair-residual tolerance.

All complex translation expectations agree with independent AO-matrix contractions within  $1.5 \times 10^{-15}$ . The GAPW comparison exposed a frozen-density handover error: keeping only the smooth real/reciprocal-space density left the new environment’s onsite hard/soft coefficients at their atomic initialization. Its quadratic spectrum differed from the original Gamma-Hamiltonian reference by  $2.67 \times 10^{-5}$  hartree<sup>2</sup> and had a true residual of  $1.71 \times 10^{-4}$  hartree<sup>2</sup>. Transferring the converged onsite coefficients before rebuilding the potential gives Table S11, without loosening tolerances or changing the independent reference.

For an exact translation by the first primitive lattice vector,  $T = SP$  is constructed independently using the cyclic cell permutation  $P$ . The complete 18-state quadratic spectrum agrees

Table S11: Periodic Gaussian translation checks.  $\Delta\lambda$  is the largest error against the independent ghost-overlap normal Gram (hartree<sup>2</sup>),  $r$  is the independently recomputed true generalized-eigenpair residual (hartree<sup>2</sup>), and  $\Delta g$  is the largest gap difference between primitive and separately converged Gamma-supercell calculations (hartree). Values are rounded upward. These are correctness checks, not scaling data.

| Cell/method | Solver         | $\Delta\lambda$       | $r$                   | $\Delta g$            |
|-------------|----------------|-----------------------|-----------------------|-----------------------|
| Cubic GPW   | Dense          | $2.2 \times 10^{-16}$ | $8.2 \times 10^{-16}$ | $2.2 \times 10^{-11}$ |
| Cubic GPW   | Iter., 2 ranks | $1.3 \times 10^{-16}$ | $3.5 \times 10^{-11}$ | $2.2 \times 10^{-11}$ |
| Skew GPW    | Dense          | $1.7 \times 10^{-16}$ | $1.2 \times 10^{-15}$ | $9.7 \times 10^{-11}$ |
| Skew GAPW   | Dense          | $7.8 \times 10^{-16}$ | $1.2 \times 10^{-15}$ | $2.0 \times 10^{-11}$ |
| Skew GAPW   | Iter., 4 ranks | $3.5 \times 10^{-16}$ | $8.7 \times 10^{-11}$ | $2.0 \times 10^{-11}$ |

within  $4.1 \times 10^{-11}$  hartree<sup>2</sup>; this comparison includes the small difference between the independent SCFs. The expectations are cube roots of unity to within  $3.0 \times 10^{-15}$ , explicitly checking the sign of the imaginary part, not just the energy. Zero-translation states give the independently evaluated  $(\epsilon_n - E)^2$  spectrum within the same bound. The largest spurious lattice-translation leakage norm is  $4.3 \times 10^{-8}$ , corresponding to squared-norm cancellation at double precision.

At a commensurate query, adding a torus lattice vector to  $\mathbf{a}$  changes the gaps by at most  $2.1 \times 10^{-15}$  hartree. Rigidly rotating cell, wave vector and translation changes them by less than  $2.3 \times 10^{-11}$  hartree. A symmetry-reduced SCF and opposite-momentum queries each agree with their respective full-SCF or opposite-query references within  $3.0 \times 10^{-15}$  hartree. These wrapping statements require commensurability:  $T_{\mathbf{a}+h_N\mathbf{n}} = T_{\mathbf{a}}$  but  $e^{i\mathbf{k}\cdot(\mathbf{a}+h_N\mathbf{n})}$  need not equal  $e^{i\mathbf{k}\cdot\mathbf{a}}$  for an arbitrary continuous query.

Additional matched dense/iterative regressions include a one-dimensionally periodic He chain and two-dimensionally periodic Ne with post-SCF GTH SOC. The latter uses TZVP-MOLOPT-GGA-GTH-q8, the SOC potential file, a  $2 \times 1 \times 1$  mesh and 68 spinor coefficients. The translations retain their open-direction components. Across the six new fixtures, all 18 requested gaps agree between serial dense and two-rank iterative runs within  $2.8 \times 10^{-14}$  hartree; the largest iterative residual is  $9.2 \times 10^{-11}$  hartree<sup>2</sup>. After formatting and rebuilding, the broader quadratic, localizer, Kubo and Wannier90 subset passes 163/163 serial and 210/210 MPI assertions. These are comparisons of the same physical operator, not an assumption that every translation preserves the AO span. Source snapshots, original AO output, states, inputs, diagnostics and a checksum manifest are retained in `validation/development/torus-translation-records.tar.gz`. The companion NumPy replay recomputes the independent checks from these records. Defective-material unfolding, spectral-weight interpretation, physical convergence and multi-node scaling are not established by this validation.

## S7.9 Silicon Bloch-band reference and integral screening

A bonded-crystal control uses diamond Si with conventional lattice constant  $a = 5.43$  Å. The primitive lattice vectors are the columns of  $h = (a/2) \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ , and the two sites are  $(a/8, a/8, a/8)$  and  $(3a/8, 3a/8, 3a/8)$ . PBE/GPW calculations use TZVP-MOLOPT-GGA-GTH-q4 and GTH-PBE-q4 from the UZH files, a 400 Ry cutoff, one grid level, relative cutoff 40 Ry, `EPS_SCF` =  $10^{-10}$  and no smearing. The SCF mesh is Gamma-centered  $3 \times 1 \times 1$ , without symmetry reduction, and the complex full-AO analysis torus has 102 scalar coefficients. These deliberately small sampling and cell choices are an algebraic validation, not a converged Si band-structure calculation.

For  $\mathbf{a} = h\mathbf{e}_1$ , the exact translation permutes cell copies. At primitive Bloch labels  $q = 0, 1/3, 2/3$ , all 34 generalized AO eigenvalues  $\epsilon_n(q)$  are obtained independently through CP2K's ordinary `BAND_STRUCTURE` output. The predicted quadratic eigenvalues at query  $k$  are the sorted union of

$$\lambda_n(q; E, k) = (\epsilon_n(q) - E)^2 + 4\kappa_T^2 \sin^2[\pi(q - k)]. \quad (\text{S7})$$

Here  $\mathbf{k} = k\mathbf{b}_1$ ,  $k = -1/2, -1/3, 0, 1/3, 1/2$ ,  $E = -0.5, -0.3, -0.1, 0, 0.1, 0.3, 0.5$  hartree and  $\kappa_T = 0.3$  hartree. The lowest three states for each of the 35 queries give 105 comparisons per solver. Continuous query labels do not add Bloch states to the three-point analysis mesh.

Table S12: Si translation analysis versus the ordinary Bloch-band reference. Errors are maxima over all 105 low eigenvalues.  $\Delta\lambda$  and the true generalized Ritz residual  $r$  are in hartree<sup>2</sup>;  $\eta_H$  is the componentwise max-entry relative Hermiticity defect. Native band energies are printed to eight decimal places in eV.

| Solver                 | $\Delta\lambda$        | $r$                    | $\eta_H$               |
|------------------------|------------------------|------------------------|------------------------|
| Serial dense           | $1.21 \times 10^{-10}$ | $7.2 \times 10^{-14}$  | $4.94 \times 10^{-14}$ |
| Iterative, 2 MPI ranks | $1.17 \times 10^{-10}$ | $9.93 \times 10^{-11}$ | $4.94 \times 10^{-14}$ |

The iterative calculation uses two OpenMP threads per rank, `MAX_SUBSPACE= 40`, `MAX_ITER= 2000` and `EPS_EIGEN= 10-10`. Across all queries, the largest dense/iterative gap difference is  $6.8 \times 10^{-12}$  hartree; the corresponding eigenvalue difference is  $5.0 \times 10^{-12}$  hartree<sup>2</sup>. The largest nominal lattice-translation leakage norm is  $9.4 \times 10^{-8}$ , corresponding to squared-norm cancellation below  $9 \times 10^{-15}$ . Individual expectations within a degenerate quadratic eigenspace can depend on the chosen basis; comparing a single state’s translation phase across solvers is not a gauge-invariant test.

An independent Gamma calculation places the six atoms at  $\mathbf{r}_A + jh\mathbf{e}_1$ ,  $j = 0, 1, 2$ , in the cell  $h \text{diag}(3, 1, 1)$ . Its ordinary Gamma Hamiltonian and overlap, rather than the lifted image matrices, define

$$Q_\Gamma = (H_\Gamma - ES_\Gamma)S_\Gamma^{-1}(H_\Gamma - ES_\Gamma) + \kappa_T^2(2S_\Gamma - z^*S_\Gamma P - zP^\dagger S_\Gamma), \quad z = e^{2\pi i k}, \quad (\text{S8})$$

where  $P$  cyclically shifts the three 34-AO cell blocks. A separate NumPy metric solve and generalized diagonalization give a maximum native-eigenvalue discrepancy of  $1.42 \times 10^{-12}$  hartree<sup>2</sup> and a largest independently recomputed Ritz residual of  $9.83 \times 10^{-11}$  hartree<sup>2</sup>. The maximum  $C^\dagger S_\Gamma C - I$  entry is  $7.3 \times 10^{-14}$ , and complex translation expectations agree with direct  $S_\Gamma P$  contractions within  $6.5 \times 10^{-14}$ . Primitive and supercell gaps agree within  $5.9 \times 10^{-12}$  hartree. Fourier-transforming the ordinary supercell matrices into the three Bloch blocks also recovers the printed primitive band energies within  $1.84 \times 10^{-10}$  hartree. Inserting these independently diagonalized blocks into Eq. (S7) reproduces the supercell quadratic eigenvalues within  $4.7 \times 10^{-12}$  hartree<sup>2</sup>.

Integral screening is controlled separately from SCF and Ritz convergence. With `EPS_DEFAULT= 10-12`, the inherited nonlocal-projector threshold is still `EPS_PPNL= 10-8`. The converged primitive Hamiltonian then has  $\eta_H = 4.114 \times 10^{-8}$ , exceeding the unchanged  $10^{-10}$  guard. Tightening `EPS_GVG_RSPACE` to  $10^{-12}$ , followed by `EPS_PGF_ORB= 10-12`, does not remove this defect. Keeping both tight values and setting `EPS_PPNL= 10-14` produces Table S12. No matrix averaging, torus-phase change or relaxed acceptance threshold is used. The implementation now reports the relative defect on failure and the largest checked Hermiticity residual on success. Focused quadratic/localizer unit and regression checks pass 30/30 serial and 54/54 two-rank MPI tests; the diagnostic also rejects a nonfinite entry owned by one rank in a four-rank column distribution.

The raw records and checksum manifest are retained in `validation/development/silicon-translation-records.tar.gz`. The replay reconstructs the Bloch-band and Gamma-matrix references. These pristine controls do not establish physical convergence.

**Displaced Si and a noncommuting translation.** Starting from the six-atom Gamma supercell above, only the second atom is moved by 0.1 Å along the Cartesian  $x$  direction. All basis, functional, screening, SCF, energy/momentum-query and iterative settings are unchanged. The displacement breaks the primitive translation, so replacing  $T$  by  $SP$  is no longer valid. A

separate Gamma overlap calculation adds translated ghost copies of all six atom-centered bases at  $\mathbf{r}_A - \mathbf{a}$ . The original/ghost off-diagonal overlap block gives  $T$  without calling the quadratic translation adapter. Connectivity generation is disabled because some ghost centers coincide with real centers; the ghost calculation is used only for overlap integrals, not for its electronic state.

The reference uses the original distorted-cell SCF  $H$  and  $S$ :

$$Q = (H - ES)S^{-1}(H - ES) + \kappa_T^2(2S - z^*T - zT^\dagger). \quad (\text{S9})$$

The independently diagonalized lowest 105 eigenvalues agree with the native two-rank iterative results within  $3.73 \times 10^{-14}$  hartree<sup>2</sup>; the largest independently recomputed Ritz residual is  $9.77 \times 10^{-11}$  hartree<sup>2</sup>. The maximum  $C^\dagger SC - I$  entry is  $1.42 \times 10^{-13}$ , complex translation means agree within  $6.0 \times 10^{-14}$ , and squared leakage norms within  $8.8 \times 10^{-14}$ . Unlike pristine Si, the metric-orthonormal Hamiltonian and compressed translation have a commutator norm of 0.3512 hartree, and  $\|S^{-1/2}(S - T^\dagger S^{-1}T)S^{-1/2}\|_2 = 0.14677$ . Across the selected states, leakage norms range from 0.003649 to 0.068246 and the largest gap change relative to the pristine cell is 0.013254 hartree. These nonzero values test the continuum projection complement and sensitivity to the distortion. They are not converged defect spectral weights or a topological classification.

### S7.10 Preserving the frozen Gamma potential

The conversion from a Gamma-only SCF environment to image-resolved operators must not retain parts of the new environment’s atomic guess. In addition to  $n(\mathbf{r})$  and its reciprocal-space grid, meta-GGAs require the converged kinetic-energy density  $\tau$ ; GAPW\_XC uses a separate soft XC density. The initializer now copies these fields, their validity information and density totals, together with the previously transferred hard/soft onsite coefficients. Allocated density gradients are transferred when valid; collocated XC gradients are not claimed to be enabled by this change.

The independent control uses He at (2, 2, 2) and (2.9, 2.1, 2.2) Å in a cubic 5 Å periodic cell, TZVP-MOLOPT-GGA-GTH-q2 and GTH-PBE-q2 from the UZH files, a 300 Ry single grid, EPS\_SCF =  $10^{-11}$  and tight orbital/projector screening. The original Gamma SCF prints its  $12 \times 12$  Hamiltonian and overlap. A separate NumPy diagonalization of  $S^{-1/2}HS^{-1/2}$  is compared with all twelve Gamma energies from the subsequent BAND\_STRUCTURE calculation. The latter exercises the conversion; the reference does not.

Table S13: Maximum post-SCF Gamma eigenvalue error relative to the original SCF matrices, in hartree. “After” is the larger error of serial and two-rank MPI runs. The native band output has eight decimal places in eV, corresponding to a rounding uncertainty of approximately  $1.84 \times 10^{-10}$  hartree.

| Method/functional | Before                 | After                  |
|-------------------|------------------------|------------------------|
| GPW/PBE           | $1.55 \times 10^{-10}$ | $1.55 \times 10^{-10}$ |
| GPW/TPSS          | $4.17 \times 10^{-3}$  | $1.82 \times 10^{-10}$ |
| GAPW/TPSS         | $8.54 \times 10^{-3}$  | $1.63 \times 10^{-10}$ |
| GAPW_XC/PBE       | $1.05 \times 10^{-2}$  | $1.83 \times 10^{-10}$ |
| GAPW_XC/TPSS      | $1.56 \times 10^{-2}$  | $1.44 \times 10^{-10}$ |

This is a shared band/frozen-potential regression, not evidence for hybrid exchange, implicit-solvent transfer or every meta-GGA property. In particular, the quadratic adapter still restricts its method selection to GPW/GAPW: the GAPW\_XC rows do not remove that guard. No Hamiltonian averaging, empirical energy shift or relaxed tolerance is used.

The paired Gamma-only/explicit-Gamma regressions pass for GPW/TPSS, GAPW/TPSS and GAPW\_XC/TPSS, with a further GAPW\_XC/PBE check. The focused quadratic and density-transfer suite passes 35/35 serial and 58/58 two-rank MPI assertions after formatting and rebuilding. Raw displaced-Si and density-transfer controls, source snapshots, scripts and checksums

are retained in `validation/development/silicon-defect-frozen-density-records.tar.gz`; the companion replay independently reconstructs the matrix references and recomputes the reported errors.

### S7.11 Stanene: a nontrivial band reference and finite-torus limits

Fresh calculations on 25 September use the two-atom stanene geometry of Section S2, but explicitly set both cell and Poisson periodicity to XY, with the analytic two-dimensional Poisson solver and 20 Å of nonperiodic cell height. This is a new boundary-condition control, not a relabelling of the earlier three-dimensionally repeated slab. The restricted PBE SCF uses a full  $8 \times 8 \times 1$  Monkhorst–Pack mesh, complex wavefunctions, threshold  $10^{-9}$ , and post-SCF GTH-PBE-q4 SOC. Two numerical settings are compared: DZVP-MOLOPT-GGA-GTH-q4 with 200 Ry and TZVP-MOLOPT-GGA-GTH-q4 with 400 Ry, both from `BASIS_MOLOPT_UZH`, with relative cutoff 40 Ry. All 26 or 34 scalar AOs are retained for the SOC diagonalization. Changing basis and cutoff together is a strengthened control, not separate extrapolations in either parameter.

For the eight occupied spinors, both Wilson surfaces converge to  $\nu = 1$  on 193 loops with 192 points per loop. The sampled indirect gaps are 0.07369133 and 0.07656851 eV. The corresponding sampled valence/conduction edges are  $(-4.36873808, -4.29504675)$  eV and  $(-4.40591061, -4.32934210)$  eV. The smallest link singular values exceed 0.95. An independent NumPy replay forms every polar-unitary link from the archived complex overlap matrices, multiplies the closed loops including reciprocal-boundary sewing, and recomputes their spectra and largest-gap crossing parity. Both parities are one; the circular WCC differences from the native spectra are below  $10^{-13}$ . Agreement on these identical matrices checks loop algebra, not an independent density-functional approximation or proof against an unsampled band crossing.

The corrected localizer uses the full AO torus at the same converged scalar potential, without occupied-band projection or spectral flattening. The query position is the Cartesian origin. Absolute query energies are  $-0.1592$  hartree for DZVP and  $-0.16050731$  hartree for TZVP, both inside the respective sampled SOC gaps. A torus of side  $N$  contains  $N^2$  primitive cells; its spinor localizer has order  $104N^2$  or  $136N^2$ , respectively. Table S14 retains the common-scale size comparison and the neighbouring-scale counterexamples. The original development scan is superseded by the SOC atom-block correction described below. Only rerun, corrected queries are used in this table; the unchanged earlier archives retain the before-fix data for provenance, not as current material-validation results.

| Basis/cutoff | $N$ | $\eta$ (hartree) | $\nu$ | Gap (hartree)            |
|--------------|-----|------------------|-------|--------------------------|
| DZVP/200 Ry  | 3   | 0.030            | 0     | $6.26487 \times 10^{-4}$ |
| DZVP/200 Ry  | 4   | 0.025            | 0     | $2.03223 \times 10^{-4}$ |
| DZVP/200 Ry  | 4   | 0.030            | 1     | $6.10518 \times 10^{-5}$ |
| DZVP/200 Ry  | 4   | 0.035            | 0     | $1.33970 \times 10^{-3}$ |
| DZVP/200 Ry  | 5   | 0.030            | 0     | $7.51780 \times 10^{-4}$ |
| DZVP/200 Ry  | 6   | 0.030            | 0     | $1.81025 \times 10^{-3}$ |
| TZVP/400 Ry  | 4   | 0.030            | 0     | $6.78326 \times 10^{-5}$ |

Table S14: Corrected finite-matrix Pfaffian indices and generalized localizer gaps from dense two-rank MPI calculations. These data supersede the original SOC torus scan and do not establish a size-, basis- and scale-converged bulk index.

At the DZVP  $N = 3$ ,  $\eta = 0.03$  query, the corrected dense gap is  $6.264872478068 \times 10^{-4}$  hartree. Independent MUMPS shifted factorizations bracket it by  $[6.264872243636, 6.264873096930] \times 10^{-4}$  hartree, and the Tacho Pfaffian path gives the same zero index, with solve-probe residual  $2.11 \times 10^{-14}$ . The serial dense gap differs by  $1.40 \times 10^{-13}$  hartree. Each MPI calculation uses two ranks and two OpenMP threads per rank, with one BLAS thread. The sparse gap test performs 66 factorizations and reports 97 MB of process-summed solver memory; this is not whole-process

memory or a performance comparison with the dense run.

The full-matrix cross-path control uses all 52 spinor eigenstates at each of the nine Gamma-centred Bloch points, together with the complete  $52 \times 52$  complex neighbour overlaps. With square coefficient matrices  $C_k^\dagger S_k C_k = I$ , their unitary polar factors are  $V_k = S_k^{1/2} C_k$ . A discrete Fourier transform of these frames places the diagonal Bloch Hamiltonian and the neighbour-link coordinate matrices into a common real-cell Löwdin basis. The independently assembled localizer is compared with  $B^{-1/2} L B^{-1/2}$  from the direct AO path. No occupied-band truncation, link unitarization or spectral flattening is applied. An orthogonal Hessenberg reduction supplies an independent real skew-matrix Pfaffian sign, checked against recursive Pfaffians on 32 small random matrices.

Before correction, the maximum localizer diagonal-block difference was  $3.10 \times 10^{-2}$  hartree, although its spin-averaged scalar part agreed within  $3.88 \times 10^{-12}$  hartree and its off-diagonal coordinate block within  $1.32 \times 10^{-15}$  hartree. The difference was traced to applying an extra atom-interchange sign to SOC image blocks whose angular-momentum integrals already include that interchange. Removing the redundant sign reduces the complete matrix difference to  $1.85 \times 10^{-12}$  hartree. The independently reconstructed gap is  $6.264872477678 \times 10^{-4}$  hartree, differing from the native value by less than  $4 \times 10^{-14}$  hartree, with the same zero Pfaffian index. This checks the torus assembly against the existing full-band SOC path, not against an independent SOC pseudopotential implementation. The Wilson calculations are unchanged.

These corrected comparisons still do not establish a nontrivial material-localizer plateau: the index changes with size, scale and the combined basis/cutoff refinement. At  $N = 3$ , the projected coordinate links have maximum unitarity defect  $\|U_j^\dagger U_j - I\|_2 = 0.4721$  and coordinate commutator norm  $\|[U_1, U_2]\|_2 = 0.3169$ . Thus the Gaussian projections cannot be treated as exact commuting unitaries at this finite basis and volume. The energy/volume conditions sufficient for the periodic localizer theorem have not been established here. A selected query matching the Wilson index is not a substitute for those checks.

The unchanged Wilson references and superseded localizer development data, with inputs, raw outputs, source snapshots and checksums, remain in `validation/development/stanene-localizer-dzvp.tar.gz` and `validation/development/stanene-localizer-tzvp.tar.gz`. The corrected runs and full-band matrix controls are retained separately in `validation/development/stanene-soc-correction.tar.gz`. Its replay verifies member checksums, reconstructs the Bloch localizer, compares the before/after AO matrices and checks the corrected dense/sparse bracket and serial/MPI agreement. A smaller  $2 \times 2$  SCF/torus regression retains all 52 spinor bands and reproduces the independently reconstructed gap to  $2.19 \times 10^{-12}$  hartree; this inexpensive fixture is not substituted for the material scan.

## S7.12 Full-band spectral flattening: size and scale controls

The next comparison retains the same scalar SCF potential, all 52 (DZVP) or 68 (TZVP) spinor bands, analytic Gaussian neighbour overlaps, query energies and Cartesian origin from Section S7.11. Only the shifted Hamiltonian is changed. In its complete Bloch eigensystem we replace each  $\epsilon_{nk} - E$  by  $\Delta \text{sign}(\epsilon_{nk} - E)$ , with  $\Delta = 1$  hartree. This retains the eight occupied spinors at every sampled k-point, their projectors, and their time-reversal structure. It is neither an occupied-only projection nor a replacement of coordinate links by their polar factors. The scale and all quoted localizer gaps below refer to this artificially flattened spectrum, not the physical SOC band gap.

At  $\eta/\Delta = 0.75$  and 1, all three larger DZVP meshes and the TZVP control agree with the nontrivial Wilson reference. This is a consistency window, not a converged localizer-gap extrapolation. At  $\eta/\Delta = 1.25$  the index is still size dependent. The  $N = 3$  control also demonstrates that flattening does not automatically repair an undersized torus. At this size we additionally retain the interpolation  $(1 - t)(\epsilon_{nk} - E) + t\Delta \text{sign}(\epsilon_{nk} - E)$  for  $t = 0, 0.5, 1$  and  $\eta/\Delta = 0.5, 1, 1.25$ . The sampled electronic gap cannot close along this interpolation, but the

| Basis/cutoff | $N$ | $\eta/\Delta$ | $\nu$ | Gap/ $\Delta$ |
|--------------|-----|---------------|-------|---------------|
| DZVP/200 Ry  | 3   | 0.50          | 1     | 0.016624      |
| DZVP/200 Ry  | 3   | 1.00          | 0     | 0.384762      |
| DZVP/200 Ry  | 3   | 1.25          | 0     | 0.618729      |
| DZVP/200 Ry  | 6   | 0.50          | 1     | 0.036234      |
| DZVP/200 Ry  | 6   | 0.75          | 1     | 0.204263      |
| DZVP/200 Ry  | 6   | 1.00          | 1     | 0.062925      |
| DZVP/200 Ry  | 6   | 1.25          | 0     | 0.079258      |
| DZVP/200 Ry  | 8   | 0.75          | 1     | 0.059190      |
| DZVP/200 Ry  | 8   | 1.00          | 1     | 0.024216      |
| DZVP/200 Ry  | 8   | 1.25          | 0     | 0.041019      |
| DZVP/200 Ry  | 9   | 0.75          | 1     | 0.263043      |
| DZVP/200 Ry  | 9   | 1.00          | 1     | 0.186946      |
| DZVP/200 Ry  | 9   | 1.25          | 1     | 0.102412      |
| TZVP/400 Ry  | 6   | 0.75          | 1     | 0.205441      |
| TZVP/400 Ry  | 6   | 1.00          | 1     | 0.064210      |

Table S15: Complete exact-sign reference scan. The  $N = 8$  control is not a multiple of three. All rows retain unmodified projected coordinate links and all spinor bands. A nonzero finite-matrix gap alone does not certify the bulk index.

$\eta/\Delta = 0.5$  finite-localizer parity changes from zero to one. A gap closing of the finite localizer must therefore occur between the sampled interpolation points.

For DZVP at  $N = 3, 6, 8, 9$ , respectively, the maximum coordinate unitarity defects are 0.47209, 0.16672, 0.10024 and 0.08074; the corresponding coordinate commutator norms are 0.31692, 0.09100, 0.05265 and 0.04182. TZVP at  $N = 6$  gives 0.25480 and 0.16651, respectively. The larger basis does not monotonically improve these operator-norm diagnostics, although the tested indices agree. The complete Fourier frames have maximum entrywise unitarity residual below  $4.8 \times 10^{-14}$ . The exact-sign operator is not strictly finite ranged, and the Gaussian-projected coordinate operators are not exact commuting unitaries. These results therefore do not establish all sufficient conditions of the periodic-localizer theorem.

The material results above are obtained independently with NumPy/SciPy from CP2K full-band exports, using the checked Hessenberg Pfaffian-sign algorithm of the preceding subsection. A separate CP2K kernel evaluates the metric-covariant sign using distributed realification, the existing Hotelling metric inverse and Newton–Schulz matrix-sign iterations. It checks inverse, sign-square, complex-structure and Hermiticity residuals rather than averaging away errors. These reference material numbers were not generated with that kernel; its subsequent native validation is reported separately below. The matrix-sign iterations avoid dense AO gathering and  $S^{-1/2}$ , but form a distributed inverse and do not by themselves guarantee limited fill-in or linear scaling. Six constructed  $12 \times 12$  complex pencils cover an orthogonal metric, nonorthogonal congruences including a more ill-conditioned metric, Kramers-paired spin mixing, and wholly positive or negative spectra. The maximum entrywise error against exact spectral constructions is below  $6.1 \times 10^{-15}$  for serial and two-/four-rank MPI runs, including row- and column-distributed block grids. Additional checks reject a zero spectral mode, the zero Hamiltonian, nonfinite query energy, a negative scale and a non-Hermitian input. The flattening, localizer and quadratic-DBCSR unit tests each pass through the regular test driver in both serial and two-rank MPI runs.

Inputs, complete exports, analysis scripts, source snapshots, constructed-matrix test logs and member checksums are retained in `validation/development/stanene-flattening.tar.gz`. The companion replay verifies all member hashes and can recompute any or all size/basis scans. Older unflattened evidence archives are unchanged.

### S7.13 Native metric-covariant flattening validation

The optional native input path applies the DBCSR sign kernel to the original AO Hamiltonian, including the full post-SCF GTH-SOC blocks when requested. Before iteration it checks the original electronic gap using the paired pencil  $\text{diag}(H - ES, -H + ES)$  with metric  $\text{diag}(S, S)$ . The paired spectrum permits reuse of the metric-aware localizer solvers; its half-signature must vanish. The dense preflight diagonalizes this pencil. The sparse preflight uses shifted MUMPS factorizations and retains a conservative gap bracket without a dense fallback. The default electronic-gap threshold is  $10^{-8}$  hartree and the matrix-sign residual threshold is  $10^{-9}$ . The four monitored residuals are the metric inverse and sign-square Frobenius residuals, and relative complex-structure and Hermiticity residuals.

For the stanene comparison we retain exactly the full-band reference geometry, scalar potential, query energies, basis/cutoff choices and unmodified coordinate links of the preceding subsection. Native two-rank DBCSR calculations agree with the independently diagonalized full-band construction, not with fitted regression values. At  $N = 6$  with DZVP, the two native gaps are 0.204263047335511 and 0.062924758531492 hartree for  $\eta/\Delta = 0.75$  and 1, respectively; both indices are one. The original electronic distance from  $E$  to the sampled spectrum is 0.001348177411 hartree. It is neither the indirect band gap nor either rescaled localizer gap. The inverse residual is  $1.67 \times 10^{-11}$  or smaller and the other three residuals are below  $8.3 \times 10^{-14}$ . The  $N = 6$  TZVP/400 Ry native control also gives two indices of one, with gaps 0.205441490284396 and 0.064209613692720 hartree. Its maximum gap difference from the independent reference is  $2.1 \times 10^{-13}$  hartree. Its inverse residual is  $5.93 \times 10^{-11}$  and the remaining residuals are below  $8.1 \times 10^{-14}$ .

| System                             | Native solver | Queries | Maximum gap difference/Ha |
|------------------------------------|---------------|---------|---------------------------|
| Stanene DZVP, $N = 3$              | Dense         | 3       | $5.7 \times 10^{-13}$     |
| Stanene DZVP, $N = 6$              | Dense         | 2       | $2.0 \times 10^{-14}$     |
| Stanene TZVP, $N = 6$              | Dense         | 2       | $2.1 \times 10^{-13}$     |
| Stanene DZVP, $N = 3$              | MUMPS/Tacho   | 2       | $7.7 \times 10^{-11}$     |
| H <sub>2</sub> , GPW               | Dense         | 8       | $2.84 \times 10^{-12}$    |
| H <sub>2</sub> , GAPW              | Dense         | 8       | $2.89 \times 10^{-12}$    |
| H <sub>2</sub> , all-electron GAPW | Dense         | 8       | $1.79 \times 10^{-12}$    |
| H <sub>2</sub> , GPW               | MUMPS         | 8       | $4.70 \times 10^{-11}$    |

Table S16: Native flattening validation against independent spectral-sign references. Sparse entries bound the difference of either returned gap-bracket endpoint (stanene) or of the reported lower bound (H<sub>2</sub>) from the reference. All stanene indices and zero molecular Chern indices agree. These are accuracy checks, not timings or material-convergence certificates.

For  $N = 3$ , the dense checks use  $\eta/\Delta = 0.5, 1, 1.25$ ; the sparse checks use 0.5 and 1. MUMPS brackets both original electronic and final localizer gaps, while Tacho evaluates the Pfaffian sign. This deliberately retains the small-volume  $\nu = 1$  to  $\nu = 0$  scale dependence instead of selecting only the nontrivial result. The independent reference lies inside both sparse localizer brackets, of widths below  $9.2 \times 10^{-11}$  hartree.

The finite checks use the UZH molecular fixtures with two energies, two positions and two localization scales. Independent SciPy diagonalization of exported  $H$ ,  $S$ ,  $X$  and  $Y$  forms  $F_E = SC \text{sign}(\epsilon - E)C^\dagger S$  for  $C^\dagger SC = I$  and  $\Delta = 1$  hartree. All 32 queries are retained, not only the final regression match. Output checks confirm two flattening operations per run, one per energy, rather than one for each of the eight localizers. Separate invalid-input runs reject an unresolved electronic gap, an insufficiently conditioned metric and a negative flattening scale before reporting an index. Constructed-matrix tests additionally reject an indefinite metric and a near-zero electronic mode; the successful six-pencil tests agree with exact spectral constructions within  $6.1 \times 10^{-15}$  in serial and on two/four MPI ranks, including a column-distributed process

grid with the sparse preflight. The final molecular/material regression directories pass 45/45 serial and 74/74 two-rank MPI assertions. The larger MPI count includes the MUMPS and Tacho directories.

Native inputs, outputs, source snapshots, basis/potential files, formatting and regression logs are retained in `validation/development/native-flattening.tar.gz`. The companion replay checks both this archive and the preceding full-band reference archive, repeats the finite-AO spectral analysis, and compares every retained native material query. Native runtime banners can predate incremental builds; the archive records the tested source snapshot separately. The larger size/scale scan in Table S15 remains an independent eigensystem study, not a claim that every row was rerun natively.

### S7.14 Complete-Bloch flattening and larger sparse torus controls

The periodic native flattening step now exploits exact translation covariance. Complete primitive Bloch eigensystems replace the whole-torus sign iteration, while the final native localizer remains in its distributed real-cell basis. Seven constructed complex pencils cover three mesh directions, one- and two-spin layouts, unequal atomic block sizes, a single-cell limit and internal degeneracy. They agree with exact matrices and the previous whole-torus route within  $3.4 \times 10^{-15}$ . Additional cases reject a broken translation, a gap closing, an indefinite overlap and a globally ill-conditioned metric even when individual Bloch metrics are well conditioned. Serial and two-/four-rank tests include different process grids. The native  $N = 3$  and  $N = 6$  stanene comparisons retain the previous independent indices, with gap differences below  $6.9 \times 10^{-13}$  hartree. The original evidence in Sec. S7.13 remains separately archived.

Profiling the larger native dense calculation also identifies a removable cubic-cost basis conversion. The signed permutation  $C$  permits the exact identity  $Q^\dagger L Q = (L + CLC + i(CL - LC))/2$ . Evaluating its four entries per output element removes the dense  $Q$ , an additional dense complex work array and their matrix multiplications. Thirteen constructed skew matrices of orders 4 through 52 reproduce both the explicit dense-unitary formula and the original matrices within  $1.4 \times 10^{-15}$ . Broken AII structure and invalid dimensions are rejected. This optimization does not change the physical time-reversal convention, factorization or gap test. The completed two-rank native DZVP  $N = 8$  calculation reproduces all three independent full-band queries within  $5.8 \times 10^{-14}$  hartree: at  $\eta/\Delta = 0.75, 1, 1.25$ , the gaps are 0.05918996709284, 0.02421594445484 and 0.04101876467039 hartree, with indices 1, 1, 0. Thus the small-torus scale-dependent counterexample is retained, not removed by the optimized conversion.

For the larger volume study, a separate sparse reference keeps the complete band basis and the original complex neighbour overlaps. In this basis the flattened Hamiltonian is diagonal, and each periodic exponential connects adjacent mesh sectors. The polar factor of each *complete MO frame* expresses an orthonormal AO frame; it does not unitarize a coordinate link. A sparse cosine/sine pairing  $W$  of  $k$  and  $-k$  is then used instead of a dense Fourier transform  $F$ . Since  $FW$  is real orthogonal, the resulting spin/mode ordering retains the same physical time-reversal involution and calibrated Pfaffian orientation as the real-space reference.

Sparse shift-invert Arnoldi iteration gives the nearest conjugate imaginary eigenpairs of the real skew representation. Eight Ritz values and their residuals against the original matrix are retained for every query. These are residual-checked spectral estimates, not shifted-inertia gap brackets. The largest residual in the retained queries is below  $10^{-8}$  hartree. Tacho independently evaluates the Pfaffian sign and must pass the same  $10^{-10}$  solve-probe tolerance as before. At most two alternative row/column permutations are tried, with their Pfaffian permutation signs included; neither the operator nor the tolerance is changed. Every failed attempt is retained. Across 15 queries at  $N = 3, 6, 8, 9$  (DZVP) and  $N = 6$  (TZVP), the sparse gaps agree with the preceding dense Fourier-frame reference within  $4.3 \times 10^{-15}$  hartree. Every resolved sparse index agrees. The  $N = 3$  cases also compare directly with dense eigenvalues and the independent Hessenberg Pfaffian in the new basis.

| Basis and torus | $\eta/\Delta = 0.75$  | 1        | 1.25     | 1.5      |
|-----------------|-----------------------|----------|----------|----------|
| DZVP, $N = 12$  | 0.291266 <sup>†</sup> | 0.229914 | 0.171938 | 0.113317 |
| DZVP, $N = 15$  | 0.306563              | 0.252124 | 0.204600 | 0.160845 |
| DZVP, $N = 18$  | 0.319876 <sup>†</sup> | 0.268152 | 0.224784 | 0.187759 |
| TZVP, $N = 12$  | 0.293046              | 0.231852 | 0.173882 | 0.115182 |

Table S17: Sparse-reference gaps of the complete-band flattened stanene localizer in hartree, with  $\Delta = 1$  hartree and the previous geometry and  $8 \times 8$  scalar SCF mesh. DZVP uses 200 Ry and  $E = -0.1592$  hartree; TZVP uses 400 Ry and  $E = -0.16050731$  hartree. All unmarked entries have resolved  $\nu = 1$  in the original dependency build. A dagger denotes an unresolved Pfaffian factorization, *not* a vanishing spectral gap or an assigned parity. The subsequent pivot-correction control below reports changed solver outcomes separately. These four sampled scales do not prove constancy between samples.

The larger torus orders are 14,976, 23,400 and 33,696 for DZVP at  $N = 12, 15, 18$ ; the  $N = 12$  TZVP order is 19,584. With DZVP at  $N = 12$ , increasing the independent scalar SCF mesh to  $12 \times 12$  changes the  $\eta/\Delta = 1$  gap by  $7.38 \times 10^{-5}$  hartree to 0.229840135569 hartree, without changing  $\nu = 1$ . The sampled electronic indirect gap changes from 0.002708107 to 0.002707169 hartree; it is not the rescaled localizer gap. At fixed SCF settings the DZVP coordinate unitarity defect decreases from 0.046929 at  $N = 12$  to 0.021374 at  $N = 18$ , and the coordinate commutator norm decreases from 0.023634 to 0.010488. TZVP at  $N = 12$  instead gives 0.084641 and 0.048762. A larger basis is therefore not a monotonic improvement of these full-space operator-norm diagnostics.

Unresolved queries from the original dependency build are deliberately retained: besides the marked table entries,  $\eta/\Delta = 0.5$  is unresolved in the sparse Pfaffian path at  $N = 3, 6, 12$ , despite residual-checked nonzero spectral gaps. The earlier dense  $N = 3, 6$  references determine their parity independently. No dense-derived index is substituted into an unresolved sparse result. The present Tacho ordering and pivoting strategy consequently remains a numerical limitation.

The larger-volume data are reference analyses of full CP2K exports, not a new sparse production representation in CP2K. Input files, complete exports, scripts, original and retried factor diagnostics, source snapshots and member checksums are retained in `validation/development/stanene-convergence/`. Its replay checks the earlier dense archive as well as the new case archives and can repeat selected sparse calculations with a Tacho-enabled CP2K shared library. Runs overlap on one machine; their elapsed times are not multi-node scaling benchmarks.

### S7.14.1 Independent pivot-correction control

A subsequent dependency-level test identifies an integer overload of the unqualified absolute-value function in the compiled skew-LDL pivot search. Subunit floating-point pivot candidates were therefore compared after truncation to integers. Replacing this call by the scalar type’s magnitude function leaves the matching, candidate rows, pivot threshold, operator entries and  $10^{-10}$  solve-residual criterion unchanged. An  $8 \times 8$  skew matrix with integer entries and exactly  $\text{Pf}(A) = 60$  supplies an independent regression: the affected library fails all six combinations of two permutations and scales 1,  $10^{-120}$ ,  $10^{120}$ , whereas the corrected library gives the expected signs with residuals below  $2 \times 10^{-15}$ . The exact value follows from recursive Pfaffian expansion, not from the corrected factorization. The small stanene  $N = 3$ ,  $\eta/\Delta = 0.5$  case then gives  $\nu = 1$  in the original ordering, agreeing with the independent dense reference. Local singular elimination blocks can nevertheless require pivot partners outside the current supernode; this correction does not implement such delayed pivots. The broader rerun also retains a new unresolved result for TZVP  $N = 12$ ,  $\eta/\Delta = 0.75$ : with the process-local BLAS, the corrected factorization exceeds  $10^{-10}$  in all three tested orderings (the smallest residual is  $1.03 \times 10^{-10}$ ). The uncorrected factorization with the same BLAS and the corrected factorization with Homebrew BLAS both resolve  $\nu = 1$ , with residuals  $8.02 \times 10^{-12}$  and  $7.81 \times 10^{-11}$ , respectively. This sensitivity is a remaining factorization

limitation, not a changed physical operator or evidence for a topology transition. No tolerance is relaxed and no failed query is replaced by a reference integer. A separate exploratory all-row maximum-column pivot policy resolves this TZVP query with a residual of  $2.25 \times 10^{-13}$ , but does not resolve the tested DZVP  $N = 6$ ,  $\eta/\Delta = 0.5$  and  $N = 12$ ,  $\eta/\Delta = 0.75$  queries. It is not substituted for the recorded production-policy results and has not completed the full validation matrix. The changed dependency, exact fixture, unchanged inputs and separate material rechecks are retained in `validation/development/pfaffian-pivot/`, without overwriting the original unresolved results.

### S7.14.2 Independent right-hand sides and symbolic-retry research

Probing a factorization only with  $b = Ax$  restricts its test right-hand sides to the range of  $A$  and can therefore conceal nullspace errors. The native adapter now solves four deterministic right-hand sides independent of  $A$  and retains the same  $10^{-10}$  relative-residual threshold. A rank-22 skew matrix of order 24, constructed as  $LDL^T$  with unit lower-triangular  $L$  and one zero  $2 \times 2$  block in  $D$ , supplies an analytic regression without zero matrix rows. Its entries are dyadic rationals, so the unscaled construction has exact nullity two. Odd permutations and scales 1,  $10^{-120}$ ,  $10^{120}$  must all be rejected. The analytic rank is used here rather than the sign of a roundoff-sized last pivot in dense elimination.

A separate deterministic set contains 600 size/scale/permutation queries at orders 8–96, including 132 numerically singular queries. The former range-restricted probes accept 72 of those singular queries; the independent probes reject all 132. For the remaining 468 queries, both versions resolve the same 450 signs correctly and leave 18 unresolved. This is a test of the factorization acceptance criterion, not evidence that the former full localizer path returned a topological index for these singular matrices: that path also requires an independently resolved spectral gap. Native unit tests pass with two and four MPI ranks; the localizer directories pass 45 dense and 29 MUMPS/Tacho checks. Three  $N = 3$  stanene controls retain their independent dense-reference gaps and parities. This subset is not a repetition of the complete 33-query material study above.

A separate, uninstalled prototype combines all-row local pivot searches with symbolic retries that move problematic matched pairs into later coupled groups. It restarts the unchanged sparse matrix instead of perturbing pivots or dropping entries. Together with the independent probes it resolves all 468 non-singular queries in the synthetic set and rejects all 132 singular queries. For DZVP stanene  $N = 6$  and  $\eta/\Delta = 0.5$ , it recovers  $\nu = 1$  with solve residual  $1.52 \times 10^{-12}$  and unchanged gap 0.036234 hartree, consistent with the retained dense reference. The accepted factorization follows 17 symbolic retries and stores 10,951,808 scalar panel entries; its largest diagonal supernode still has order 2,950 out of 3,744. These are panel counts, not total process-memory measurements. A preceding  $N = 12$  experiment was interrupted after more than 100 retries without an accepted result. Thus restart-based coarsening demonstrates a possible recovery mechanism but does not establish a scalable delayed-pivot solver and is not enabled in CP2K. The checks, source snapshots and unsuccessful trace are retained separately in `validation/development/pfaffian-probes/`.

### S7.14.3 Delayed skew pivots without symbolic restarts

A subsequent, separate numerical extension passes rejected pivot variables and their Schur contributions to the parent front of the Tacho elimination tree. Original matrix entries are assembled exactly once, at the front owning their earlier symbolic endpoint. A child update contains both its separator and any delayed variables. A candidate pivot is tested against both complete updated columns, including the separator, with relative threshold 0.01 and the unchanged absolute cutoff on the normalized matrix. At the root a rook search replaces further delays. No matrix entries are dropped and no pivot perturbation or reference-sign substitution is used.

For an accepted skew block  $D_j = \begin{pmatrix} 0 & d_j \\ -d_j & 0 \end{pmatrix}$ , the multipliers are  $L_{21} = A_{21}D_j^{-1}$  and the trailing update is  $A_{22} - L_{21}D_jL_{21}^T$ . A blocked implementation accumulates 64 accepted columns before applying this update with BLAS level 3. The sign is the product of the signs of the upper-block entries  $d_j$ , multiplied by the parity of the complete elimination permutation. Stored panels also provide forward, block-diagonal and backward solves for the four independent right-hand sides.

At this prototype checkpoint the implementation is an uninstalled, serial skew-specific extension, not a change to the general indefinite solver or a distributed Pfaffian factorization. Its final research adapter uses the public Tacho `GraphTools_Metis` and `SymbolicTools` interfaces; private driver accessors were used only in preceding experiments. For accepted test components of order at most 512, an additional dense check reconstructs  $LDL^T$  in the actual elimination order. This checks the factors; it is not a dense fallback for the index.

The 600-query synthetic suite now resolves all 468 nonsingular queries and rejects all 132 singular queries. An additional deterministic set contains 636 queries, with orders 2–258, four random permutations per matrix, three overall scales, banded, separator, dense and disconnected patterns, and analytically singular dyadic congruences. All 336 nonsingular queries are resolved correctly and all 300 singular queries are rejected. These tests run with undefined-behaviour instrumentation of the adapter and new numerical extension and C++ standard-library assertions; no instrumentation error is reported. The separately compiled dependencies are not covered by that instrumentation. Across these two sets, 834 accepted nontrivial component factors are explicitly reconstructed, with maximum relative Frobenius residual  $1.04 \times 10^{-13}$ . Rebuilding the adapter from its standalone CMake entry point and replaying both sets reproduces the same signs and singular rejections.

The complete 33-query stanene study is then repeated from the unchanged, hash-verified exports, with the archived basis-specific query energies. All 33 queries are resolved in the first external ordering; the five previously unresolved queries in the floating-pivot-corrected scan are listed in Table S18. All 28 previously resolved parities are retained, and all 15 available independent dense references agree, with maximum gap difference  $4.24 \times 10^{-15}$  hartree. The gaps remain independent near-zero Ritz estimates checked against the original matrix, not rigorous shifted-inertia brackets. The small-torus trivial indices are retained as well: the improved factorization does not force a nontrivial classification.

| Torus/basis   | $\eta/\Delta$ | Gap (hartree) | $\nu$ | Solve residual         |
|---------------|---------------|---------------|-------|------------------------|
| 6 × 6, DZVP   | 0.50          | 0.03623405    | 1     | $1.14 \times 10^{-12}$ |
| 12 × 12, DZVP | 0.50          | 0.01628035    | 1     | $2.06 \times 10^{-11}$ |
| 12 × 12, DZVP | 0.75          | 0.29126552    | 1     | $2.18 \times 10^{-13}$ |
| 18 × 18, DZVP | 0.75          | 0.31987626    | 1     | $2.88 \times 10^{-13}$ |
| 12 × 12, TZVP | 0.75          | 0.29304631    | 1     | $5.82 \times 10^{-13}$ |

Table S18: Previously unresolved queries recovered by the separate delayed-skew prototype. The operator and the  $10^{-10}$  acceptance threshold are unchanged. The residual is the worst relative two-norm residual of four independent right-hand sides, not an estimate inferred from pivot magnitudes.

For  $N = 12$ ,  $\eta/\Delta = 0.5$ , the order-14,976 problem uses 84 fronts (largest 4,582), 46,049,224 scalar panel entries and 15,804 transfers between ancestors, not distinct delayed variables. For  $N = 18$ ,  $\eta/\Delta = 0.75$ , the order is 33,696, the largest front is 5,408 and the panel count is 77,431,958. The four-query  $N = 18$  process peaks at 10.52 GB, including Python complete-band and sparse-matrix containers. This is not factor storage alone or a distributed-memory scaling result. A boundary test rejects a 676-million-entry front before allocation against the 650-million-entry array limit. This cap excludes maps, right-hand sides and other workspaces.

The numerical extension, public-interface build, sanitizer checks, complete material rerun and independent archive replay are retained in `validation/development/pfaffian-delayed/`.

Earlier unresolved results are preserved rather than overwritten. The following subsection records native integration separately; a whole-workflow resource policy and MPI-distributed factors remain separate requirements.

#### S7.14.4 Native integration and structural-roundoff controls

The delayed-skew numerical extension is subsequently integrated into CP2K’s existing narrow C interface, using the public Tacho symbolic analysis and BLAS interfaces. It does not copy the dependency’s fixed-front numerical implementation or introduce a general-purpose indefinite solver. The input keyword `PFAFFIAN_MEMORY` limits requested double-precision factor, pivot, frontal, Schur-update and solve buffers in MiB, with default 4096. The matching, sparse input, symbolic ordering, integer metadata, allocator overhead, DBCSR and MUMPS are outside this limit. The reported peak covers only these numerical buffers, not whole-process memory. Scalar panel counts are reported separately. Disconnected components are processed sequentially, so their peaks form a maximum while their panel counts are summed. Tests exercise an exactly sufficient budget, a one-byte reduction, collective exhaustion and inconsistent positive limits across MPI ranks.

The first complete integration run exposed a Bismuth SOC failure that the exactly skew model matrices did not reveal. Tolerated floating-point roundoff can give a nonzero entry on only one side of the skew matrix; passing that asymmetric graph to symbolic analysis can omit a required frontal entry. The native implementation consistently uses

$$A_s = \frac{1}{2}(\text{Re } A - (\text{Re } A)^T)$$

and the symmetric union of the sparse patterns, without an entry-drop threshold. The discarded part  $E = A - A_s$ , including its imaginary part, obeys

$$\|E\|_2 \leq \sqrt{\|E\|_1 \|E\|_\infty} \leq \max(\|E\|_1, \|E\|_\infty).$$

This computed upper bound must pass both the structural tolerance and the existing gap-relative bound in the positive AO metric. It is not legitimate to restore physical time reversal when this test fails. A 32-site skew-chain reproducer tests all 930 possible one-sided off-pattern couplings of magnitude  $10^{-15}$ . Before correction, 435 placements fail; afterwards all signs agree with the exact chain Pfaffian. Transposed, odd-permuted and extreme-scale variants enter the native unit test, and larger skew defects remain rejected. The Bismuth SOC gap is 0.0681101299687 hartree, consistent with the unchanged dense reference, with solve residual  $1.40 \times 10^{-15}$ .

The native C++ source also passes the 1,236-query synthetic suite under undefined-behaviour instrumentation and standard-library assertions: 804 nonsingular queries have the expected signs and 432 singular queries are rejected. Prebuilt dependencies are not instrumented. The finite and periodic CP2K localizer directories pass 74 checks with four MPI ranks and two OpenMP threads; a build without the optional sparse libraries passes all 45 dense checks. These are integration tests, not a distributed Pfaffian scaling measurement. The first failed run, its reduced reproducer, corrected outputs and source provenance are retained in `validation/development/pfaffian-native/`.

Finally, the native shared library repeats all 33 complete-band material queries from the unchanged exports and the basis-specific archived energies. Every query is resolved in the first external ordering, with unchanged parity and maximum independent-solve residual  $4.07 \times 10^{-11}$  at the  $10^{-10}$  threshold. The native path uses the default public ordering options, so factor counts and residuals need not equal the prototype’s. The 15 available dense references are retained in the independently verified reference archive. This is a validation of the native factorization using the complete-band representation, not a new native large-torus SCF calculation or a replacement of the Ritz gap estimates by inertia brackets.

#### S7.15 Separating basis, cutoff, SCF sampling and cell height

The preceding DZVP/200 Ry and TZVP/400 Ry comparison changes two numerical parameters at once. A separate control series fixes the analysis torus at  $12 \times 12$  while varying the Gaussian basis

and density-grid cutoff independently. All scalar states enter the post-SCF SOC diagonalization, and all resulting spinors enter the full-band reference localizer. No empty-band truncation or unitarization of the coordinate links is introduced. The query energy is placed at the midpoint of each sampled global electronic gap. For the complete-band sign flattening this does not change the operator when the energy remains within the same gap. Three independent scales,  $\eta/\Delta = 0.75, 1, 1.25$ , are tested with  $\Delta = 1$  hartree.

Fresh DZVP/200 Ry and TZVP/400 Ry SCF calculations first reproduce the earlier reference controls. Within DZVP, changing the cutoff from 200 to 400 Ry changes the  $\eta/\Delta = 1$  localizer gap by  $4.35 \times 10^{-7}$  hartree; increasing it further to 600 Ry changes the gap by  $4.05 \times 10^{-7}$  hartree in the opposite direction. These small, nonmonotonic differences are retained rather than interpreted as an extrapolation law. At the common 400 Ry cutoff, the DZVP-to-TZVP change is  $1.94 \times 10^{-3}$  hartree, which separates the basis effect from the much smaller cutoff effect.

| Basis/cutoff/SCF      | $E_g$ (meV) | $g_{0.75}$ | $g_1$    | $g_{1.25}$ |
|-----------------------|-------------|------------|----------|------------|
| DZVP/200/8            | 73.691341   | 0.291266   | 0.229914 | 0.171938   |
| DZVP/400/8            | 73.690847   | 0.291265   | 0.229913 | 0.171937   |
| DZVP/600/8            | 73.691304   | 0.291265   | 0.229914 | 0.171938   |
| TZVP/400/8            | 76.568521   | 0.293046   | 0.231852 | 0.173882   |
| TZVP/600/8            | 76.568488   | 0.293046   | 0.231853 | 0.173882   |
| TZVP/600/12           | 76.541027   | 0.292849   | 0.231779 | 0.173788   |
| TZVP/600/18           | 76.535992   | 0.292874   | 0.231786 | 0.173796   |
| TZVP/600/12, $R = 60$ | 76.541029   | 0.292849   | 0.231779 | 0.173788   |
| TZVP/600/12, $h = 25$ | 76.540999   | 0.292849   | 0.231779 | 0.173788   |

Table S19: Separately varied numerical controls for complete-band stanene on a fixed  $12 \times 12$  analysis torus. Cutoffs are in Ry; the final number specifies the square scalar SCF mesh. Unless marked otherwise, the relative cutoff  $R = 40$  Ry and open-axis cell height  $h = 20$  Å. The buckling is fixed at  $0.852$  Å.  $E_g$  is the sampled electronic indirect gap;  $g_r$  is the localizer gap in hartree at  $\eta/\Delta = r$ , with  $\Delta = 1$  hartree. All listed indices are  $\nu = 1$ . The rescaled localizer gaps are not electronic band gaps.

The further controls vary the scalar SCF integration mesh, the relative grid cutoff and the open-axis cell height independently. When the height changes, the Cartesian buckling remains  $0.852$  Å; only the corresponding fractional coordinates change. The square property mesh and its reciprocal-boundary neighbour list are independent of the scalar SCF integration mesh. Automated input checks verify the reciprocal lattice, every mesh step and boundary wrap, and the fixed Cartesian geometry. The reference analyses run in separate processes with one BLAS thread; the preceding CP2K exports use two MPI ranks and two OpenMP threads per rank. Peak resident memory of the analysis process is distinguished from the native numerical-buffer limit and from the earlier MPI export.

With TZVP/600 Ry, changing the scalar SCF mesh from  $8 \times 8$  to  $12 \times 12$  changes the  $\eta/\Delta = 1$  gap by  $-7.34 \times 10^{-5}$  hartree; the subsequent  $18 \times 18$  control changes it by  $6.91 \times 10^{-6}$  hartree. Increasing the relative cutoff from 40 to 60 Ry changes the gap by  $1.20 \times 10^{-9}$  hartree. Increasing the open-axis height from 20 to 25 Å changes it by  $-2.24 \times 10^{-8}$  hartree. All 27 queries in these nine fixed-volume controls give  $\nu = 1$ . The electronic gap changes from 76.5685 to 76.5410 and then 76.5360 meV along the three TZVP/600 Ry SCF meshes; this is not the roughly 0.232-hartree gap of the flattened localizer.

A separate Wilson surface for the TZVP/600 Ry,  $12 \times 12$  SCF potential converges to  $\nu = 1$  after four refinements, with a sampled electronic gap of 76.5410 meV. Its final surface contains 193 loops with 192 points per loop. The largest reported WCC change is  $2.62 \times 10^{-4}$  at the final refinement. The two coarsest candidate parities are zero and are explicitly not accepted as converged invariants. This independent check underscores the difference between a gap-resolved Hamiltonian and adequately sampled Wilson transport.

The independent DZVP/200 Ry,  $8 \times 8$ -SCF volume control extends the analysis torus to  $21 \times 21$ , giving a localizer of order 45,864. At  $\eta/\Delta = 0.75, 1, 1.25$ , the gaps are 0.331864624081, 0.282051686763 and 0.240476515333 hartree, respectively, and all three indices are one. In particular, the  $\eta/\Delta = 1$  gap remains 5.2% above the  $18 \times 18$  value. This is evidence for consistency of the sampled index, not quantitative volume convergence of the gap. The coordinate unitarity defect and commutator norm decrease to 0.015787 and 0.007697, respectively; neither is set to zero by modifying the coordinate operators. The largest analysis-process resident-memory high-water mark is 9.05 GiB for this control. A dense complex matrix of this order alone would occupy 31.3 GiB before eigensolver workspaces. These measurements concern the complete-band reference representation, not MPI-distributed Pfaffian factors or a controlled timing benchmark.

All 30 new complete-band queries resolve the Pfaffian in the first external ordering at the unchanged  $10^{-10}$  solve-probe threshold. The new records, including full exports, method snapshots and an archive-verification and numerical-replay script, are retained in `validation/development/stanene-separated-controls/`. The table and pairwise differences are regenerated directly from the retained reports rather than transcribed from run summaries.

### S7.16 Bi(111) periodic reference for finite-flake validation

The fixed Bi(111) geometry uses an in-plane lattice constant of 4.33 Å and a vertical sublattice separation of 1.74 Å, following Li et al., *New J. Phys.* **23**, 063042 (2021), [doi:10.1088/1367-2630/ac04c9](https://doi.org/10.1088/1367-2630/ac04c9). The primitive cell has two Bi atoms, a 20 Å open-axis height, and identical PERIODIC XY settings for the cell and Poisson solver. Restricted PBE, the UZH DZVP-MOLOPT-GGA-GTH-q5 basis, GTH-SOC pseudopotentials, a 400 Ry cutoff and 40 Ry relative cutoff define this test. Its complex full-grid  $8 \times 8 \times 1$  scalar SCF calculation converges in 75 iterations to  $-11.059654777634860$  hartree. All 26 scalar AOs enter the post-SCF SOC second variation; the selected Wilson space contains the lowest ten spinors, corresponding to ten valence electrons.

Starting with 13 Wilson loops of 12 points each, the adaptive calculation retains  $\nu = 1$  through three refinements and meets the  $10^{-3}$  WCC-change tolerance. The final surface contains 97 loops with 96 points per loop, with maximum reported WCC change  $5.5218 \times 10^{-4}$  and transverse step 0.019109. The sampled indirect electronic gaps are 0.405115, 0.403917, 0.403222 and 0.403202 eV on successive surfaces. The complete SCF-plus-Wilson run takes 75.93 s with two MPI ranks and two OpenMP threads per rank on the local validation machine. This is a single-run duration, not a scaling result. The inputs, full outputs, eigenvalues and overlap links are retained in `validation/development/bismuth-finite-validation/`. The original per-run driver is archived rather than retrospectively replaced by its later version. This result establishes convergence of the Wilson sampling for this fixed Hamiltonian, not complete-basis convergence or a topological classification of an edge-relaxed flake.

### S7.17 Finite Bi patch and one-image SOC storage

The finite counterpart is a bare, unrelaxed  $3 \times 3$  patch of the two-atom Bi geometry, with 18 atoms and no edge passivation. A total padding of 20 Å is added to each Cartesian extent, with matching PERIODIC NONE settings and the MT Poisson solver. The restricted PBE calculation uses the same UZH DZVP and GTH-SOC data, 400/40 Ry cutoffs, a  $10^{-9}$  SCF threshold and 300 K Fermi–Dirac smearing. All 234 scalar AOs and 468 post-SCF spinors are retained. The complex localizer has order 936, or 1872 after realification.

The reference independently integrates the Gaussian AO overlap and first moments from the exported primitive contractions, and reconstructs the full covariant spinor Hamiltonian from the complete eigenframe. Its overlap ratio is 0.00453546, the frame metric residual is  $1.18 \times 10^{-13}$ , and the time-reversal residual is  $4.45 \times 10^{-15}$ . The three directly printed finite AO SOC components agree with those extracted from this Hamiltonian to  $7.91 \times 10^{-16}$  hartree, with largest metric-normalized error  $3.97 \times 10^{-15}$  hartree.

This comparison first failed for the two-rank export although the native dense and sparse localizers agreed. A two-atom reproducer showed correct onsite SOC blocks but missing interatomic blocks; the one-rank calculation agreed with the direct finite operator. The cause was selecting matrix storage from the number of cell images rather than the actual k-point mode. Even with only one image, the real-to-reciprocal transform requires triangular storage on the neighbour-list owners. Desymmetrizing beforehand redistributed the blocks and made local lookups miss them. The corrected two-rank reproducer matches all finite AO SOC components to  $2.91 \times 10^{-16}$  hartree. Independent diagonalization of printed  $H, S, V_x, V_y, V_z$  also reproduces the separate GPW/GAPW regression example's occupied eigenvalues to  $3.2 \times 10^{-14}$  eV and its gap of 1.057726627969 eV, including a four-rank check. Both row-labelled and `OMIT_HEADERS` matrix printing are exercised. The broader localizer/topology suites pass 86 MPI and 57 serial numerical checks.

The retained flake scan uses absolute energies  $-0.1588789410086769$ ,  $-0.1552040084684156$ , and  $-0.1515290759281543$  hartree; scales  $\kappa = 0.001, 0.003, 0.01$  hartree/bohr; and positions at the centroid and at Cartesian  $x$  offsets of 6 and 12 Å. These fixed diagnostic energies are not defined by the corrected neutral gap midpoint. All 27 native and independent indices agree: five are nontrivial and 22 trivial. The independently evaluated localizer gaps span  $5.06009 \times 10^{-4}$  to 0.117384 hartree. Their maximum distance outside the native shifted-inertia brackets is  $2.72 \times 10^{-12}$  hartree, at an unchanged comparison tolerance of  $2 \times 10^{-8}$  hartree. The largest reported Pfaffian solve residual is  $1.35 \times 10^{-13}$ ; numerical Pfaffian buffers occupy 13.829 MiB. This buffer figure is not total process memory and does not establish distributed Pfaffian scaling.

The restarted calculation, 27 native queries and complete export take 269.35 s on two MPI ranks with two OpenMP threads per rank; the independent dense analysis takes 4.53 s. These durations cover different work and are not a solver speedup comparison. The corrected finite neutral electronic gap is only  $2.84260 \times 10^{-4}$  hartree, smaller than the smearing scale. Consequently, neither a finite nontrivial query nor the agreement of algorithms certifies a bulk insulating phase. Size, edge, basis and temperature convergence remain necessary. The full corrected flake record, the deliberately retained failing two-rank dimer diagnostic, its one-rank control, the corrected two-rank counterpart, source patches and numerical verification methods are archived in the same Bi validation directory.

## S7.18 Neutral-gap and frontier-state diagnostics

To compare finite patches without choosing unrelated absolute energies, a separate independent scan sets  $E = E_{\text{mid}} + f g_{\text{el}}$ , where  $g_{\text{el}}$  is the neutral post-SCF SOC gap and  $f = -1/4, 0, 1/4$ . Query positions are the centroid plus Cartesian  $x$  displacements of 0, 0.75, and 1.5 times the atomic half-width in that direction. These are geometric sampling positions, not an orbital-support boundary. The scales are  $10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}$ , and  $10^{-2}$  hartree/bohr. The unchanged 18-atom, DZVP, 300 K snapshot gives eight nontrivial and 37 trivial queries. All nontrivial points occur at the centroid; even there the index is not constant over the sampled scales. This independent scan is distinct from the 27 like-for-like native comparisons above. Archive-only replay reproduces all 45 gaps, indices and frontier weights without numerical change. Passing the same reconstructed operators to the native Tacho C interface also reproduces all 45 indices, with largest solve-probe residual  $5.40 \times 10^{-12}$ . This additional comparison checks factorization, not native AO assembly or a second SCF calculation.

A complementary diagnostic resolves the frontier eigenspaces over atoms. For a complete degenerate group  $\mathcal{B}$ , the reported weight is

$$w_A(\mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \sum_{\sigma, \mu \in A} |(S^{1/2} C_{\sigma})_{\mu n}|^2. \quad (\text{S10})$$

It is invariant under unitary rotations within  $\mathcal{B}$ , but retains the usual basis dependence of Löwdin atom populations. A 3.3 Å bond cutoff separates the 3.043 Å nearest neighbours from the 4.33 Å second neighbours. Ten of 18 atoms have fewer than three neighbours and are designated as

boundary atoms. The occupied and unoccupied frontier Kramers doublets place 72.29% and 79.65% of their respective weights on these atoms, compared with a boundary atom fraction of 55.56%. The small finite gap,  $g_{\text{el}}/(k_B T) = 0.2992$  at 300 K, therefore comes with substantial boundary weight. This is a diagnostic of this finite patch, not proof of topological edge transport or of a converged insulating bulk gap. Analytic tests check the subspace weights under degenerate rotations and in a nonorthogonal AO metric. Here  $T$  controls the scalar-reference SCF smearing, not ionic motion or self-consistent spinor occupations; SOC is added after the restricted SCF calculation.

Lowering the scalar SCF smearing to 30 K, with the same geometry, basis and cutoffs, requires 168 SCF steps from the retained 300 K checkpoint. The subsequent independent scan gives three nontrivial and 42 trivial queries: all three nontrivial cases lie at the centroid with  $\kappa = 0.003$  hartree/bohr, one at each selected energy. The neutral electronic gap decreases rather than opening upon cooling the scalar reference (Table S20). The frontier boundary weights remain high. Thus this temperature comparison does not establish convergence or remove the need for edge and size controls.

Table S20: The same bare 18-atom Bi patch at two scalar SCF smearing temperatures. Frontier weights average each complete Kramers pair. The boundary contains 10 of 18 atoms in both calculations.

| $T$ (K) | $g_{\text{el}}$ (meV) | $g_{\text{el}}/k_B T$ | Occupied edge (%) | Empty edge (%) |
|---------|-----------------------|-----------------------|-------------------|----------------|
| 300     | 7.735                 | 0.2992                | 72.29             | 79.65          |
| 30      | 2.178                 | 0.8424                | 72.55             | 80.12          |

An additional native 30 K calculation restarts from this converged spectrum and converges in seven SCF steps. Its 45 queries use exactly the energies, positions and scales of the independent 30 K scan; the dense reference is reconstructed from the new run’s own complete export. All indices agree, with maximum distance outside the native gap brackets of  $6.56 \times 10^{-11}$  hartree at the unchanged  $2 \times 10^{-8}$  hartree comparison tolerance. The largest native Pfaffian solve residual is  $6.15 \times 10^{-12}$ . The full restarted calculation, queries and export take 371.28 s on one MPI process and two OpenMP threads with one BLAS thread. One-second samples record a maximum process-tree RSS sum of 19.6 GiB over 365 samples with no failed observations. This is a sampled resident-memory sum, which may count shared pages repeatedly, not an exact peak, unique physical memory or Pfaffian-buffer size. The preceding 168-step SCF/export took 2167.64 s and did not have RSS telemetry.

### S7.19 A continuous finite-model scale window

The index need not remain constant for arbitrarily small or large  $\kappa$ . At the centroid and each calculation’s own neutral gap midpoint, a finer scan identifies a common nontrivial window near 0.003 hartree/bohr. To check the unsampled intervals, write

$$L(\kappa) = L(0) + \kappa D, \quad \|D\| = \|(X - xI) - i(Y - yI)\| = 19.89764 \text{ bohr}. \quad (\text{S11})$$

Here  $X, Y$  are the complete, metric-orthonormalized finite AO position matrices. The smallest absolute eigenvalue obeys  $|g(\kappa) - g(\kappa')| \leq \|D\| |\kappa - \kappa'|$ . For a subinterval  $[a, b]$  with midpoint  $m$ , the implementation therefore uses the lower bound

$$g(\kappa) \geq g(m) - \|D\|(b - a)/2 - \epsilon_{\text{num}}, \quad \epsilon_{\text{num}} = 10^{-10} \text{ hartree}. \quad (\text{S12})$$

Recursive bisection covers  $[0.0025, 0.0035]$  hartree/bohr with 11 accepted subintervals and 21 midpoint evaluations for each temperature. The minimum covered bounds are  $9.10 \times 10^{-5}$  hartree at 300 K and  $7.95 \times 10^{-5}$  hartree at 30 K. Since the anchor at  $\kappa = 0.003$  has  $\nu = 1$ , neither finite model can change its index within this numerically resolved interval. The explicit roundoff margin

is not an interval-arithmetic proof, and the conclusion applies at the specified energy and position, not simultaneously to all sampled queries or to the infinite material.

The original methods, native outputs, spectra and moments, neutral scans, factorization comparison and interval bounds are retained under `validation/development/` in the `bismuth-neutral-validation/` directory, together with links to the earlier 300 K evidence. Archive-only replay of the 30 K scan independently checks all 45 queries and frontier weights. Analytic tests include invariant degenerate-subspace weights, gap-cover tiling and rejection of unresolved or nonfinite gaps. These controls are numerical validation, not substitutes for physical convergence of the flake.

## S7.20 A controlled change of finite termination

The original parallelogram patch has two atoms with only one neighbour within 3.3 Å. A separate neutral calculation removes these two atoms while preserving the simulation cell and the coordinates of all 16 remaining atoms. No passivation, structural relaxation or artificial change to the Hamiltonian matrix is used. The same PBE, UZH DZVP, 400/40 Ry grids and 300 K scalar smearing now describe 80 electrons, 208 scalar AOs and all 416 spinors. The calculation converges in 131 SCF iterations from the atomic guess. Its post-SCF SOC gap is 0.008071023745 hartree (219.6 meV), with  $g_{\text{el}}/(k_B T) = 8.4954$ , compared with 7.735 meV for the 18-atom termination at the same temperature. The occupied and empty frontier doublets have boundary weights 82.62% and 79.70%, respectively. Ten of 16 atoms are boundary atoms (62.5%); strong frontier boundary weight therefore persists even though the electronic gap is much larger.

The neutral-window scan has three nontrivial and 42 trivial queries, with the nontrivial cases at the centroid and  $\kappa = 0.003$  hartree/bohr. The off-centre sampling is rescaled to the new patch half-width, so query counts alone are not a like-for-like comparison between terminations. At the centroid and neutral midpoint, the 16-atom localizer instead changes its index between  $\kappa = 0.0025$  and 0.00275 hartree/bohr. The attempted gap cover over  $[0.0025, 0.0035]$  therefore correctly fails: it retains an unresolved region near 0.0026215985 hartree/bohr after 249 midpoint evaluations. This negative result is archived, not replaced by a successful narrower scan. Varying  $\kappa$  changes the localizer, not the fixed electronic Hamiltonian or its electronic gap.

For  $[0.00275, 0.0035]$  hartree/bohr, 13 midpoint evaluations cover seven subintervals with  $\nu = 1$ , a norm constant of 15.22720 bohr and a minimum numerical lower bound of  $4.64539 \times 10^{-5}$  hartree. This interval is also contained in the resolved intervals of both 18-atom temperature controls. Thus all three finite Hamiltonians have a common central scale window, although this does not remove their geometry and basis dependence or establish a thermodynamic limit.

A five-step native restart evaluates the same 45 queries and exports a fresh complete spectrum for independent reconstruction. All 45 indices agree, with maximum distance outside the native gap brackets  $3.48 \times 10^{-10}$  hartree and largest Pfaffian solve residual  $2.36 \times 10^{-13}$ . The comparison tolerance remains  $2 \times 10^{-8}$  hartree. This native SCF/query/export run takes 301.50 s; the independent dense analysis takes 4.99 s. Its sampled process-tree RSS maximum is 18.5 GiB over 296 successful one-second observations. These timings cover different tasks and must not be interpreted as a factorization speedup.

The 131-step scalar SCF and complete export take 1794.77 s on one MPI process and two OpenMP threads. The 1751 one-second RSS observations have no failures and give a maximum process-tree sum of 16.5 GiB, with the same memory-accounting caveats as above. The workstation is an Apple M4 Max with 36 GiB RAM; the retained host record also gives OS and numerical-library versions. Raw termination inputs, outputs, methods, complete exports, successful and rejected interval bounds are retained separately in `bismuth-termination-validation/` under the development validation directory.

## S7.21 A physical cross-basis comparison of the finite Bi states

The 16-atom termination is also calculated with UZH TZVP at the same positions, cell, 300 K scalar smearing, grid cutoffs and post-SCF SOC potential. The converged DZVP restart supplies only an initial scalar guess; all 272 scalar AOs and 544 spinor states are recomputed. The SCF converges in 119 steps, and the full run takes 1628.81 s. Its 1588 one-second memory samples have no failures and give a process-tree RSS maximum of 16.4 GiB. The electronic gap becomes 0.008515556694 hartree (231.7 meV), 5.51% above DZVP. The occupied and empty frontier boundary weights are 80.66% and 77.19%, compared with 82.62% and 79.70% in DZVP. Such Lowdin weights themselves depend on the AO basis and are not the basis-convergence metric used below.

To compare actual states across bases, the reference independently integrates the rectangular physical overlap  $S_{ab}^{\text{DT}} = \langle \phi_a^{\text{D}} | \phi_b^{\text{T}} \rangle$  from the exported Gaussian contractions. No coefficient padding, matching of individual degenerate eigenvectors or identification of separate Lowdin coordinate frames is used. For any selected spinor subspaces, the singular values of

$$M = C_{\text{D}}^{\dagger} \text{diag}(S^{\text{DT}}, S^{\text{DT}}) C_{\text{T}} \quad (\text{S13})$$

are their principal-angle cosines. Each frame is first checked in its own physical metric. Complete occupied manifolds and complete frontier Kramers pairs are compared. For equal ranks,  $\|P_{\text{D}} - P_{\text{T}}\| = \sqrt{1 - \sigma_{\min}(M)^2}$  and  $\|P_{\text{D}} - P_{\text{T}}\|_F^2 = 2r - 2\sum_j \sigma_j(M)^2$ . Tests verify invariance under independent unitary frame rotations and nonsingular AO basis changes, including complex matrices, and reject non-normalized or nonfinite inputs.

The DZVP AO space is embedded in TZVP using  $E = (S^{\text{TT}})^{-1} S^{\text{TD}}$ . In the DZVP orthonormal metric, the residual of  $S^{\text{DD}} - E^{\dagger} S^{\text{TT}} E$  is  $2.86 \times 10^{-14}$ ; all three corresponding position-operator residuals are below  $1.17 \times 10^{-12}$  bohr. This checks the physical nested basis and its operators, not just matrix dimensions. The self-consistent Hamiltonians, however, are different: their projected difference has norm 0.0053733 hartree. This is a basis-dependent change of the SCF potential, not an export error.

For the 80-dimensional occupied spaces, the smallest overlap singular value is 0.99449263, the projector distance is 0.10480657, and the mean retained weight is 99.6866%. The corresponding projector distances for the occupied and empty frontier doublets are 0.09276768 and 0.06853998. A same-export DZVP control reproduces its Hamiltonian within  $4.51 \times 10^{-14}$  hartree; principal-angle sines near  $4 \times 10^{-8}$  in that control reflect the square-root sensitivity of  $1 - \sigma^2$  to roundoff.

Of the same 45 normalized position–energy–scale queries, two are nontrivial in TZVP instead of three in DZVP. At the centroid and  $\kappa = 0.003$  hartree/bohr, the lower-quarter neutral-gap query becomes trivial while the midpoint and upper-quarter queries remain nontrivial. The neutral midpoint itself changes with basis, so these energies are defined relative to each calculation’s own gap, not held at a common absolute value. The attempted bound on  $[0.00275, 0.0035]$  fails after 271 evaluations near 0.002851703 hartree/bohr and is retained as a negative result. The smaller interval  $[0.003, 0.0035]$  is resolved with  $\nu = 1$ , norm constant 16.04838 bohr and minimum lower bound  $1.12998 \times 10^{-4}$  hartree. This interval is contained in all previously resolved windows, but the shift of its lower boundary and the cross-basis subspace distances prevent a claim of complete-basis convergence.

A four-step native TZVP restart evaluates the same 45 queries and exports its own complete spectrum. All indices match the independent reconstruction, including the lower-quarter query that differs from DZVP. The maximum gap distance outside the native bracket is  $5.21 \times 10^{-10}$  hartree, at the unchanged  $2 \times 10^{-8}$  tolerance, and the largest Pfaffian solve residual is  $1.08 \times 10^{-13}$ . The full native run takes 486.42 s, compared with 9.49 s for its independent dense analysis; these include different work and do not measure a solver speedup. Its 478 successful one-second samples give a process-tree RSS maximum of 20.4 GiB. The fresh-export electronic gap changes by only  $4.02 \times 10^{-11}$  hartree from the spectrum run.

## S7.22 A separate finite-size control

Increasing the patch from three to four primitive cells in each in-plane direction and removing the two singly coordinated corners gives 30 atoms, 150 electrons, 390 UZH DZVP AOs and 780 spinors. The 300 K smearing, 400/40 Ry grids, 20 Å vacuum and SOC potential are unchanged. The bounding cell grows with the bare, unrelaxed flake. The atomic-guess SCF converges in 132 steps; SCF and export take 2601.85 s on the same one-rank, two-thread setup. The 2530 successful one-second samples give a process-tree RSS maximum of 19.4 GiB, with the qualifications above.

The neutral SOC gap is 0.002650895533 hartree (72.14 meV), compared with 219.6 meV for the smaller DZVP patch, and  $g_{\text{el}}/(k_B T) = 2.7903$ . Fourteen of 30 atoms are boundary atoms, whereas the complete occupied and empty frontier doublets carry 80.82% and 71.04% of their Lowdin weight on the boundary. The declining electronic gap and strong boundary weights prevent an interpretation as a converged bulk band gap.

The original 45-query scan has three nontrivial results, all at the centroid and  $\kappa = 0.0003$  hartree/bohr. This equals the smaller DZVP patch's count but at a tenfold smaller scale; equal counts obscure the physical change. A 20-scale scan at the neutral midpoint and centroid finds nontrivial samples at 0.00025–0.0004 and 0.0015–0.00275 hartree/bohr, separated by trivial samples. The coarse grid missed the latter region.

Numerical gap covers confirm two nontrivial intervals rather than interpolating these sampled indices. On [0.00025, 0.0004] hartree/bohr, 17 midpoint evaluations cover nine subintervals with minimum lower bound  $1.82336 \times 10^{-5}$  hartree. On [0.0015, 0.0025], 15 evaluations cover eight intervals with minimum lower bound  $7.67239 \times 10^{-5}$  hartree. Both use the same norm constant, 22.28339 bohr, and  $10^{-10}$  hartree numerical margin. In contrast, the interval [0.003, 0.0035], which is nontrivial in both 16-atom bases, is entirely trivial for the 30-atom Hamiltonian: nine evaluations cover five intervals with minimum lower bound  $8.73676 \times 10^{-5}$  hartree and anchor  $\nu = 0$ . This is a resolved opposite index, not a failed or unresolved numerical factorization. These finite-size changes must remain visible when evaluating possible material convergence.

A separate Tacho C-interface check of all 20 reconstructed fine-scan skew matrices matches every Pfaffian index, with maximum solve residual  $2.11 \times 10^{-12}$ . This tests factorization, not native AO assembly, and does not replace the following check.

The separate five-step native restart evaluates all 45 original queries and exports a fresh complete spectrum. Every native index matches its independently reconstructed counterpart, with maximum distance outside the native gap bracket  $1.31 \times 10^{-10}$  hartree and largest Pfaffian solve residual  $2.02 \times 10^{-13}$ . The comparison tolerance remains  $2 \times 10^{-8}$  hartree. The full native run takes 1169.99 s; the separate dense reconstruction and analysis take 24.31 s. The 1147 successful one-second samples give a process-tree RSS maximum of 18.2 GiB, not a factorization-only memory measurement. The spectrum-run and native-run electronic gaps differ by  $3.62 \times 10^{-11}$  hartree.

Cases, original methods, positive and negative bounds, cross-basis comparisons and tests are retained under `validation/development/` in `bismuth-size-basis-validation/`, without replacing the earlier termination and temperature records. Archive-only replay verifies 270 members in six archives and reproduces both 45-query scans, the physical cross-basis comparison and all reported accepted or rejected interval bounds without numerical change. This replay is separate from the native runs.

## S8 Independent Wilson and TRIM Eigenframe Symmetry

This validation concerns an independent post-SCF property point set, not reuse of SCF quadrature weights. The default full-diagonalization route is compared with the optional K290 and SPGLIB property backends. All calculations retain the original ordered links and reciprocal boundary shifts. The new regression inputs use MOLOPT UZH bases; SOC cases use GTH SOC pseudopotentials.

The scalar He fixture has one atom at fractional coordinates (1/2, 1/2, 1/2) in a cubic 4 Å cell. Its four-point directed NNKP loop includes an explicit reciprocal-cell closing shift. Both

backends use three scalar diagonalizations and recover the full-loop Berry phase  $\pi$ . For Ne/SOC in a cubic 5 Å cell, the refined  $8 \times 5$  property surface has 40 points, reduced to 15 scalar and 15 SOC diagonalizations. All 13 scalar AO states are retained in the second variation, and the exported subspace contains eight occupied spinors. All five Wilson spectra and 40 directed links are compared, not just the trivial final index. An earlier five-scalar-state window cuts a degenerate unoccupied group. That input is retained as an expected rejection, rather than used as a quantitative reference.

The comparison evaluator independently integrates contracted Cartesian Gaussians with the exported spherical contraction coefficients. For every point it obtains the physical sewing matrix  $G = C_{\text{full}}^\dagger S C_{\text{rec}}$  and tests both frame norms,  $G^\dagger G - I$ , and the residual subspace norm of  $C_{\text{rec}} - C_{\text{full}} G$ . Every directed MMN connection is checked against  $G_i^\dagger M_{ij}^{\text{full}} G_j$ . Wilson centers are compared modulo one with permutation and branch-cut handling. A fixed  $2 \times 10^{-7}$  threshold is used for these development comparisons, including energies in hartree; the Ne and He errors are orders of magnitude smaller. Seven synthetic sensitivity tests include complex degenerate spinor gauges in a nonorthogonal metric, missing subspace components, wrong metrics, wrong link conjugation, branch cuts, and nonfinite overlap data.

The nontrivial stanene TRIM control uses the geometry of the earlier parity fixture, two Sn atoms and 26 scalar AO functions. Its 52-state second-variational problem exports eight occupied spinors. A Gamma-centered, symmetry-reduced  $2 \times 2 \times 1$  SCF mesh with 400 Ry cutoff, 100 Ry relative cutoff, and  $10^{-10}$  SCF tolerance gives two property representatives for the four TRIM. K290 and SPGLIB both retain  $\nu = 1$ . For SPGLIB, independent comparison with four full diagonalizations gives a maximum complete-spectrum discrepancy  $3.76 \times 10^{-13}$  hartree, a physical subspace residual  $1.68 \times 10^{-13}$ , and an MMN covariance residual below  $1.6 \times 10^{-14}$ . TRIM connections in this test are self-links; they do not establish Wilson-surface convergence. A separate  $4 \times 4 \times 1$ , 1200/100 Ry control gives complete-spectrum discrepancies below  $3.2 \times 10^{-13}$  hartree for both backends.

The same 400/100 Ry frozen-potential setup also tests nontrivial Wilson connections. An initial  $8 \times 5$  surface with one allowed refinement reaches 144 points and nine loops, with 30 representatives. Full diagonalization, K290 and SPGLIB all report insufficient Wilson sampling convergence. Nevertheless, the exported frames and connections agree: the SPGLIB full-spectrum error is below  $3.5 \times 10^{-13}$  hartree, the physical subspace residual below  $3.1 \times 10^{-13}$ , and the directed-link covariance residual below  $2.2 \times 10^{-13}$ . This validates reconstruction, not the convergence of that coarse surface.

A second calculation starts with 12 points per loop and 13 transverse loops and permits four refinements. All three routes terminate normally with  $\nu = 1$  at 192 points per loop and 193 loops, or 37,056 points. The successive representative counts are 61/156, 217/600, 817/2352, 3169/9312 and 12,481/37,056 for both backends. The final comparison checks every occupied eigenvalue, every Wilson spectrum modulo one, and the singular spectra of all 37,056 directed MMN links. For K290 and SPGLIB, respectively, the maximum occupied-energy errors are  $8.50 \times 10^{-13}$  and  $2.05 \times 10^{-12}$  eV, the Wilson-center errors are  $2.21 \times 10^{-13}$  and  $2.10 \times 10^{-13}$ , and the all-link singular-value errors are  $4.92 \times 10^{-13}$  and  $4.27 \times 10^{-13}$ . This large run omits the full AO-state export; its external comparison is gauge-invariant, whereas the smaller 144-point run additionally verifies physical subspaces and the full gauge-covariant link matrices. Native H/S and SOC residual checks remain active in both runs.

These are correctness measurements, not a demonstrated runtime gain. Single complete runs, including SCF and all refinements, take 226.1 s without property reduction, 254.3 s with K290 and 252.5 s with SPGLIB on an Apple M4 Max using two MPI ranks and two OpenMP threads per rank, with one BLAS thread. In this small 26-AO problem, orbit planning, AO-frame reconstruction and target-operator checks offset the saved scalar and SOC diagonalizations; cross-k integrals and links are still built explicitly. Repeated timings, larger bases and distributed property representatives would be needed for a scaling claim.

The rejected controls are significant. The original shifted, unreduced  $4 \times 4 \times 1$  SCF mesh at

200/40 Ry breaks threefold equivalence: the maximum spectral splitting among the nominally equivalent M points is  $1.06127 \times 10^{-3}$  hartree. Both reconstruction backends reject its frozen H/S problem. Symmetry-preserving SCF sampling alone is insufficient: at 600/40 Ry an independent comparison still found  $2.86206 \times 10^{-7}$  hartree error in the full spectrum. This motivated the additional projected-H residual check. Raising the relative integration cutoff, rather than relaxing an acceptance threshold, eliminates that discrepancy in the tested controls. These are operator-consistency checks, not a complete-basis or physical material-convergence claim.

Ordinary incomplete SCHEME GENERAL integration stars remain invalid. Separate negative runs check this for K290 and SPGLIB, the scalar-window degeneracy boundary for both backends, and an invalid operator tolerance. The property-only planner may handle incomplete stars, but cannot change the SCF closure requirement. A multi-atom test also exercises complete mirrored image-cell storage for the dense Fourier H/S check; the sparse source retains its original triangular DBCSR storage convention.

The final native regression driver passes 220 checks in each of the SSMP and two-rank PSMP runs, spanning the new 12-input directory and the existing topology, little-group and Kubo directories. Two OpenMP threads and one BLAS thread are used, with an explicit 64 MiB OpenMP stack. An earlier serial attempt using the runtime-default stack timed out during pre-property grid collocation and was stopped; its log and stack sample are retained and are not counted as a successful run. No numerical reference or operator tolerance was changed for the completed repeat.

The automatic TQC smoke uses two Ne atoms, 16 occupied spinors, 56 sampled points and 82 directed compatibility segments. All three routes remain compatible and preserve the sampled atomic signature. This particular generated graph already has 56 independent property points, so it tests interface compatibility rather than additional eigenproblem reduction. The final build also repeats the symmetry-breaking stanene control: full diagonalization completes, whereas both reduced routes reject the fourth point with AO-scaled H residual  $1.2923 \times 10^{-5}$  and projected-H residual  $2.8702 \times 10^{-4}$ . The metric residual is only  $1.69 \times 10^{-14}$ , demonstrating why normalization alone is insufficient.

Inputs, final native sources, build settings, rejected controls and comparison methods are retained in `validation/development/property-symmetry/`. Archive-only replay verifies 394 file entries in 28 archives and passes all 23 quantitative comparisons and seven sensitivity tests. The frame comparisons include direct SSMP-PSMP full-diagonalization controls as well as both reconstruction backends. Across these comparisons, the largest full-spectrum discrepancy is  $8.24 \times 10^{-13}$  hartree, the largest physical subspace residual  $3.08 \times 10^{-13}$ , and the largest directed-link covariance residual  $2.12 \times 10^{-13}$ . The two large-surface comparisons use the separate gauge-invariant criteria described above. This replay reruns the archived comparison algorithms, not the native DFT calculations. Final shared-library checksums are recorded along with executable checksums; the earlier exploratory runs retain their original, executable-only provenance.

## S8.1 Independent scalar MPI groups

A subsequent validation distributes the independent scalar diagonalizations over CP2K's existing k-point MPI groups. The tested group sizes are automatic, all ranks in one group, and two ranks per group, with two and four total MPI ranks and two OpenMP threads per rank. Both K290 and SPGLIB are compared with full diagonalization for He, Ne/SOC and stanene/SOC. The He case deliberately distributes three representatives over two two-rank groups. The final Ne mesh distributes 15 representatives over four one-rank groups. These unequal workloads test local indexing and transfer from the owning group, rather than only the evenly divisible case. Native checks require exactly one owning group for each representative.

All 24 physical frame/link comparisons pass. Across the matrix, the maximum complete-spectrum error is  $7.23 \times 10^{-13}$  hartree, the maximum physical subspace residual  $1.69 \times 10^{-13}$ , and the maximum directed-link covariance residual  $1.94 \times 10^{-14}$ . The largest Wilson-center difference

is  $8.89 \times 10^{-16}$  in reduced units. Four invalid group requests are rejected, and four additional four-rank runs reject a scalar SOC window ending inside a degenerate group, for both backends and both one- and two-rank groups. The latter checks that the additional boundary eigenvalue is communicated consistently. There are 35 native cases in total, including these expected failures. The regular 13-input property directory also passes 27 checks in SSMP and 27 with four-rank PSMP.

The records in `validation/development/property-groups/` include native inputs and outputs, source and shared-library provenance, grouping diagnostics, and the independent comparison programs. Archive replay repeats the 24 physical comparisons without a DFT rerun. This validates parallel correctness, not multi-node performance. At this checkpoint, SOC assembly, reconstruction checks, and directed links were not distributed across the independent k-point groups. The earlier 37,056-point timings above belong to the preceding single-group implementation.

## S8.2 SOC orbit assembly in independent MPI groups

The next stage assigns complete SOC orbits to the same groups that own their scalar representatives. Each group Fourier-transforms bounded snapshots of the three imaginary real-space SOC operators, projects them into its reconstructed scalar frames, diagonalizes the representatives, and validates every partner against its own target SOC Hamiltonian. Conjugating mirrored blocks includes the imaginary-operator factor, rather than treating SOC as a real scalar operator. An ownership reduction checks that every requested target contributes exactly once to the collected spectra and spinors. This does not reduce the directed-link graph.

In addition to the preceding He, Ne/SOC and stanene/TRIM cases, the comparison uses three explicitly specified eight-point stanene loops starting at fractional coordinates  $(1/8, 1/8, 0)$ ,  $(1/8, 3/8, 0)$  and  $(1/8, 5/8, 0)$ , with a reciprocal shift at each closure. Their 24 points reduce to six representatives. Four one-rank groups therefore have unequal orbit workloads. These non-TRIM, multiatom tests check complex phases and directed connections at fixed resolution; they are not an additional claim of Wilson-surface convergence or of a converged material invariant.

The full matrix compares both backends using two and four MPI ranks, automatic grouping, a single group, and two ranks per group, with two OpenMP threads and one BLAS thread. All 32 comparisons pass. The largest complete-spectrum difference is  $4.96 \times 10^{-13}$  hartree, the largest physical subspace residual  $1.69 \times 10^{-13}$ , and the largest directed-link covariance error  $1.44 \times 10^{-13}$ . Wilson centers agree to  $3.26 \times 10^{-14}$  in reduced units. These comparisons use independently integrated Gaussian overlaps, not only the code's internal normalization checks. Group diagnostics confirm that the SOC stage uses the requested scalar grouping in every reduced run; none enters the collective snapshot fallback.

Including four invalid group requests and four degenerate-window rejections, the matrix contains 44 native cases. The property directory passes all 29 checks with SSMP and four-rank PSMP, including the two new non-TRIM stanene loops. The existing `QS/regtest-kp-3` directory passes another 16 checks in each build, covering ordinary scalar uses of the shared Fourier-operator service as well as forces and legacy/general k-point inputs. No existing numerical reference, operator tolerance or Wilson convergence tolerance was changed for these results. This establishes local parallel and gauge correctness; it is not a measurement of multi-node scaling.

The archive in `validation/development/property-soc-groups/` retains the native inputs and outputs, operator sources, build configuration, executable and shared-library checksums, and comparison methods. Archive-only replay verifies all 463 member files and repeats the 32 physical comparisons with the same maximal discrepancies. It does not rerun the DFT calculations. At this checkpoint, scalar target-frame validation and directed-link construction remained collective stages; their independent distribution was not yet established.

### S8.3 Group-local scalar target checks

A further stage distributes the scalar target-frame validation, not only representative diagonalization and SOC assembly. Each group processes the targets of its representatives using bounded real-space H/S snapshots, rather than dense arrays of every real-space image on every rank. The three physical residual tests are shared with the previous collective implementation. A stable orbit ordering and a source-frame cache avoid retaining all target frames. Each round transfers at most one checked frame per group into the existing full-graph MO storage, with exactly one delivery per point. Idle groups participate in the same transfer sequence.

The additional all-electron GAPW control uses He with the UZH TZVPP-MOLOPT-GGA-ae basis, a 400 Ry cutoff and a 100 Ry relative cutoff. All 14 scalar orbitals, including p/d components and degenerate spaces, are retained in the frame/link comparison. Four explicit loop points reduce to three independent eigenproblems. The full-diagonalization reference uses two MPI ranks; K290 and SPGLIB each use two one-rank or two two-rank groups. These are operator and parallel-consistency tests, not a complete-basis or material-convergence study.

The final He/Ne/stanene matrix passes all 32 physical frame/link comparisons with group-local scalar checks enabled. The largest complete-spectrum discrepancy is  $5.96 \times 10^{-13}$  hartree, the largest physical subspace residual  $1.73 \times 10^{-13}$ , and the largest directed-link covariance residual  $1.73 \times 10^{-13}$ . Wilson centers agree to  $3.47 \times 10^{-14}$  in reduced units. Every reduced run reports the expected validation grouping, without the collective H/S snapshot fallback.

All four GAPW comparisons also pass. Their largest complete-spectrum difference is  $1.25 \times 10^{-14}$  hartree, the largest physical subspace residual  $4.59 \times 10^{-15}$ , and the largest directed-link covariance error  $1.20 \times 10^{-14}$ . Wilson centers differ by at most  $6.67 \times 10^{-16}$  in reduced units. Together with the He/Ne/stanene matrix, this gives 36 independent physical comparisons.

Four additional four-rank tests reject a scalar SOC window cutting a degeneracy. The asymmetric-SCF stanene control is also repeated on four ranks: full diagonalization completes, while both reduced routes reject target point four. Its metric error remains below  $3.0 \times 10^{-14}$ , but the AO-scaled and projected-Hamiltonian errors are  $1.2923 \times 10^{-5}$  and  $2.8702 \times 10^{-4}$ . Thus a successful norm check cannot conceal the physically invalid orbit. These expected failures complete collectively, without leaving an MPI group waiting for a missing transfer.

Including the four invalid group requests, the final matrix contains 52 native cases. The 17-input property directory passes all 33 checks in SSMP and four-rank PSMP; the shared-operator directory QS/regtest-kp-3 passes another 16 checks in each build. All runs use two OpenMP threads and one BLAS thread, without changing existing reference values or physical acceptance tolerances.

The retained evidence in `validation/development/property-target-groups/` contains the native inputs and outputs, final sources, build logs, executable and shared-library checksums, and comparison methods. Archive-only replay verifies all 541 payload files and passes the 36 physical comparisons without rerunning DFT. Directed-link construction still uses the complete MPI environment; its independent distribution and multi-node scaling are not established by these tests.

## S9 Submission Validation Still Required

The current record does not establish a nontrivial, size-converged DFT localizer classification for a large SOC flake, multi-node sparse scaling, or general 230-space-group TQC. Further SOC systems and converged flakes require basis/cutoff controls, spatial/energy/scale scans, independent band references, timings and whole-process memory. DZVP/TZVP is not a complete-basis extrapolation, and development diagnostics or plans must not be reported as completed material validation.